The Rise of Inaction: Job Reallocation and the Declining Labor Share*

Byunghee Choi[a], Choongryul Yang[b]

[a]*Mount St. Mary's University*
[b]*Federal Reserve Bank of Atlanta*

Abstract

Since 2000, the U.S. labor share and job reallocation rate have both fallen. We document a concurrent rise in establishment inaction and a shift toward smaller employment adjustments. Industry-level data across U.S. states show a positive association between reallocation and payroll shares, with larger increases in establishment inaction accompanying larger declines in both. A general equilibrium model of heterogeneous firms with fixed, kinked, and quadratic adjustment costs provides a common explanation. Higher costs lower establishment value and, through free entry, reduce the equilibrium wage relative to output per worker. Across two targeting exercises, rising costs account for 24–37% of the labor-share decline without targeting it, while generating increased inaction and a larger share of small employment adjustments.

Keywords: Labor share, Job reallocation, Business dynamism, Labor market frictions

JEL classification: E24, E25, J23, J42

1. Introduction

Since 2000, the U.S. economy has experienced concurrent declines in the labor share and business dynamism. Between 2001 and 2019, the labor share fell from about 57% to 53%, while the annual job reallocation rate—the sum of job creation and destruction rates—fell from 31% to 22%. This paper asks whether rising labor adjustment costs can help explain both trends. The connection runs through the equilibrium wage: costlier future adjustment lowers the value of operating an

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Email addresses: b.choi@msmary.edu (Byunghee Choi), choongryul.yang@atl.frb.org (Choongryul Yang)

establishment at a given wage. At the initial wage, the expected value of entry falls below the entry cost. A lower wage restores free entry and reduces the labor share.

We first document the establishment behavior underlying declining reallocation. Public Business Employment Dynamics (BED) data show that the share of establishments with exactly zero net employment change in a quarter rose from 45.5% in 2001 to 52.0% in 2019. Among establishments that expand, the share adding only 1–4 jobs also rose, from 82.4% to 85.5%. Thus, declining reallocation combines more inaction with smaller adjustments among establishments that adjust. These two margins jointly discipline the adjustment technology.

We then connect establishment inaction to reallocation and labor shares in a state-industry panel. Using Census Business Dynamics Statistics (BDS) and BEA data, we find that job reallocation and payroll shares move together over time within a given industry and state, even after controlling for market concentration, unionization, and manufacturing exposure. A bridge analysis using BED data shows that larger increases in establishment inaction accompanied larger declines in BDS reallocation and payroll shares. In the panel regressions, the relationship appears both at continuing establishments and in establishment entry and exit, and it is larger for entry and exit.

Guided by these facts, we build a general equilibrium model of heterogeneous establishments in the tradition of Hopenhayn and Rogerson (1993). Establishments face persistent and transitory productivity shocks and pay fixed, kinked, and quadratic employment adjustment costs. We calibrate the benchmark to ten 2001 moments, including count-based and employment-weighted quarterly inaction and three points of the conditional gains-size distribution. We then increase the adjustment cost parameters to match the observed rise in quarterly inaction and decline in quarterly gross job flows. The labor share falls by 0.95 percentage points and annual reallocation by 5.88 points, accounting for about 24% and 68% of their observed declines. The increase in the share of small adjustments also emerges without being targeted in the counterfactual.

Across the baseline and an alternative targeting exercise, rising adjustment costs account for 24–37% of the labor-share decline. In the alternative, we increase the kink and quadratic cost parameters proportionally to match the full annual reallocation decline. The labor share then falls by 1.47 percentage points, or 37% of its observed decline. The labor-share response is untargeted in both exercises. The difference reflects whether the cost increase is disciplined by quarterly establishment behavior or by the full annual reallocation decline.

The cost components play complementary roles. The kink helps reproduce the inaction response, while the quadratic component makes large changes progressively more costly. Their joint increase matches the changes in inaction and gross flows and raises the share of small adjustments. Increasing the fixed and quadratic costs instead gives a similar aggregate result. The comparisons support a rise in the cost of large adjustments, while allowing more than one combination of cost increases to explain the two quarterly changes.

Our model extends the analysis of declining business dynamism by linking establishment adjustment behavior to the labor share. The panel evidence connects the aggregate trends to changes within industries in individual states, and quarterly BED inaction and gross flows discipline the baseline cost increase. The declines in the labor share and annual reallocation and the rise in the conditional small-adjustment share are then untargeted predictions whose directions and magnitudes can be compared with the data. Studying the two aggregate trends in one framework requires the account of declining reallocation to agree with evidence on factor shares and adjustment sizes. These additional predictions provide a basis for evaluating the common mechanism.

The wage response is central because a reduction in reallocation alone need not lower the labor share. Higher adjustment costs reduce establishment value at the initial wage, inducing a lower wage under free entry. They also change the allocation of employment and reduce output per worker. The latter effect partly offsets the wage contribution to the labor-share decline. Quantifying both effects is therefore necessary to predict how declining responsiveness changes factor shares.

Our contribution builds on the evidence of declining firm responsiveness in Decker et al. (2020). They distinguish changes in productivity shocks from changes in firms' responses to those shocks. We document the connection to inaction using public data and extend their analysis to factor shares. Under the same inaction and gross-flow targets, both rising costs and smaller shocks reduce reallocation. Rising costs lower the labor share, while smaller shocks raise it. The labor-share response therefore distinguishes the two mechanisms in this comparison.

This mechanism complements explanations of the labor-share decline based on capital substitution, market power, and firm composition. Karabarbounis and Neiman (2014) emphasize investment prices and capital substitution. Glover and Short (2020) and Oberfield and Raval (2021) examine the relevant substitution elasticities. Autor et al. (2020), De Loecker et al. (2020), and Barkai (2020) study concentration, markups, and profits, while Böckerman and Maliranta (2012)

and Kehrig and Vincent (2021) distinguish within- and between-firm movements in labor shares. Our general equilibrium analysis also relates to the role of allocation in Baqaee and Farhi (2020).[1] Our approach adds a link from the frequency and size of employment changes to factor shares.

The business-dynamism literature documents declines in job and worker flows (Davis et al., 2012; Hyatt and Spletzer, 2013) and examines explanations involving mobility and institutions (Molloy et al., 2016), knowledge diffusion (Akcigit and Ates, 2023), short employment spells (Pries and Rogerson, 2022), and housing and credit conditions (Davis and Haltiwanger, 2024). Albrecht and Decker (2026) find no systematic industry-level association between rising markups and declining dynamism. We contribute a common adjustment-cost mechanism whose implications for both dynamism and the labor share are disciplined by establishment behavior.

The model draws on the adjustment-cost literature surveyed by Hamermesh and Pfann (1996), including the role of inaction in Bloom (2009) and plant-level labor adjustment in Lee and Mukoyama (2018) and Cooper et al. (2015). Cooper et al. (2024) examine rising adjustment costs and declining responsiveness. In the labor-search literature, Cooper et al. (2007) combine fixed vacancy-posting costs and variable adjustment costs to explain establishment behavior and aggregate fluctuations. We follow their approach of using the frequency and size of establishment adjustments to discipline aggregate implications. Fujita and Ramey (2007) study sunk costs of creating job positions that remain available, filled or vacant, until obsolete. Increasing marginal creation costs encourage gradual creation, and durable positions generate persistent vacancy responses. Our quadratic term uses the related mechanism of rising marginal costs to limit employment changes. Related models with multi-worker firms include Elsby and Michaels (2013), Kaas and Kircher (2015), and Schaal (2017). We extend this literature in two respects: we use secular adjustment patterns to evaluate which cost increases explain declining dynamism, and we quantify their untargeted labor-share implications through the free-entry wage response. Section 6.2 connects the cost comparisons to these mechanisms.

Section 2 describes the data, and Section 3 presents the panel evidence. Section 4 documents establishment inaction and adjustment sizes, links inaction to the state-industry panel, and reviews

[1] See Elsby et al. (2013) and Grossman and Oberfield (2022) for surveys. Related work examines labor market power (Berger et al., 2022; Yeh et al., 2022; Mertens and Schoefer, 2024; Gouin-Bonenfant, 2022), firm heterogeneity and organization (Hubmer and Restrepo, 2026; Smith et al., 2022), and time-series evidence (Bergholt et al., 2022).

evidence on rising adjustment costs. Sections 5 and 6 develop the model and quantitative results. Section 7 concludes.

2. Data

We construct a state-by-industry panel that combines data on labor income shares, job reallocation, and market concentration from several administrative and survey sources. Our main sample covers 2001–2019, the benchmark-to-counterfactual period studied below, and ends before the COVID-19 pandemic, which introduced extreme disruptions to employment dynamics. We discuss each data source in turn. Additional detail on data construction is provided in Appendix A.

2.1. Labor Income Share

Our measure of the labor income share is the *payroll share*, following the methodology of Elsby et al. (2013) and Karabarbounis (2024). For each state-industry-year cell, we define:

$$\lambda_{ist} = \frac{\text{Compensation of Employees}_{ist}}{\text{Gross Value Added}_{ist}} \tag{1}$$

where i indexes industry, s indexes state, and t indexes year. The numerator is total compensation of employees (wages, salaries, and supplements), and the denominator is gross domestic product by state (value added). Both are sourced from the Bureau of Economic Analysis (BEA) Regional Economic Accounts. These data are available annually at the state-by-industry level for broad NAICS-based sectors. We map these to 11 private-sector industry groups that correspond to the industry classifications available in the Census Business Dynamics Statistics.[2]

2.2. Job Reallocation

We construct measures of labor market fluidity from the Business Dynamics Statistics (BDS), produced by the Census Bureau from the Longitudinal Business Database. The BDS is an annual dataset that tracks job flows built from establishment-level employment changes—job creation from expanding and opening establishments, and job destruction from contracting and closing establishments—at the state-by-sector level.

[2]By construction, this payroll share excludes proprietors' income, which avoids the well-known ambiguity of allocating self-employment income between labor and capital (Elsby et al., 2013; Grossman and Oberfield, 2022).

Following Davis et al. (1998), we define job creation (*JC*) as the sum of all employment gains at expanding and opening establishments and job destruction (*JD*) as the sum of all employment losses at contracting and closing establishments. These gross flows are normalized by average employment to yield the Job Creation Rate (*JCR*) and Job Destruction Rate (*JDR*). Our primary measure of labor market fluidity is the Job Reallocation Rate (*JRR*):

$$JRR_{ist} = JCR_{ist} + JDR_{ist} \tag{2}$$

which captures the total "churn" or dynamism in the labor market. The BDS reports data at the state-by-sector level using 2-digit NAICS codes, which we aggregate to 11 private-sector industry groups matching the BEA classification (see Appendix A.1 for the sector mapping). For the establishment-level size-of-change analysis in Section 4.1, we use the BLS Business Employment Dynamics (BED), which provides data on the size of establishment-level employment changes.

2.3. Market Concentration

To control for the product-market-power channel, we construct an employment-based Herfindahl-Hirschman Index (HHI) at the state-by-industry-year level using the Statistics of U.S. Businesses (SUSB) from the Census Bureau. The SUSB reports, for each state-industry-year cell, the number of firms (N^k_{ist}) and total employment (E^k_{ist}) in each firm-size class k. Because the SUSB reports firm sizes in bins, we approximate the firm-level HHI by assuming equal-sized firms within each bin:

$$HHI_{ist} = \sum_k \frac{1}{N^k_{ist}} \left(\frac{E^k_{ist}}{E_{ist}} \right)^2 \tag{3}$$

where $E_{ist} = \sum_k E^k_{ist}$ is total employment in the cell and the division by N^k_{ist} converts each bin's squared employment share into the sum of squared shares of the individual firms in that bin. Higher HHI values indicate greater employment concentration.

2.4. Additional Controls: Unionization and Manufacturing Share

To control for alternative explanations of the labor share decline, we construct two additional state-level variables. First, we measure *private-sector union membership rates* from the database of Hirsch and Macpherson (2003), which provides state-by-year union density estimates derived

from the Current Population Survey. Declining unionization has been widely cited as a contributing factor to the falling labor share (Stansbury and Summers, 2020), and we use union rates to assess whether our JRR–labor-share finding is absorbed by this measured channel. Second, we compute the *manufacturing employment share* at the state level using the SUSB data. We include the manufacturing share to control for measured changes in state industrial composition.[3]

2.5. *Establishment-Level Size-of-Change Data*

We use BLS data on establishments classified by the *size* of their quarterly employment change. The BLS BED program reports, for the aggregate private sector, the number of establishments and associated employment flows broken down by the magnitude of their employment change: 1–4 jobs, 5–19 jobs, 20 or more jobs, and *zero change*. We compute the inaction rate as the fraction of all establishments reporting zero net employment change in a given quarter.[4]

2.6. *Analysis Panel*

We merge these data sources into a state-by-industry-year panel. The unit of observation is a state (s) $\times$ industry (i) $\times$ year (t) cell, where industries correspond to 11 private-sector groups. The panel also includes state-level union membership rates and manufacturing employment shares as additional controls. After restricting to cells with non-missing observations for the payroll share, job reallocation rate, and HHI, our main estimation sample contains 10,631 state-industry-year observations spanning 561 state-industry cells over 2001–2019.

Appendix Table A.1 presents the summary statistics in three panels. Panel A reports the pooled distribution: the average payroll share is 56.6% with substantial cross-sectional variation (standard deviation of 16.9%), and the average JRR is 27.5%. Private-sector union membership averages 6.6% across states and years, while manufacturing accounts for 11% of state employment on average. Panel B documents the secular trends: between 2001 and 2019, the mean payroll share fell from 60.2% to 55.4%, the JRR declined from 31.6% to 22.7%, and union membership fell from 8.2% to 5.9%. The grouped-data HHI rises modestly over the sample and remains small in level terms. It

[3] Elsby et al. (2013) distinguish historical composition effects from the more recent decline, which mainly reflects falling payroll shares within industries.

[4] Zero change means exact equality of reported employment in the third month of consecutive quarters. Offsetting hires and separations or adjustments reversed between observations can leave employment unchanged.

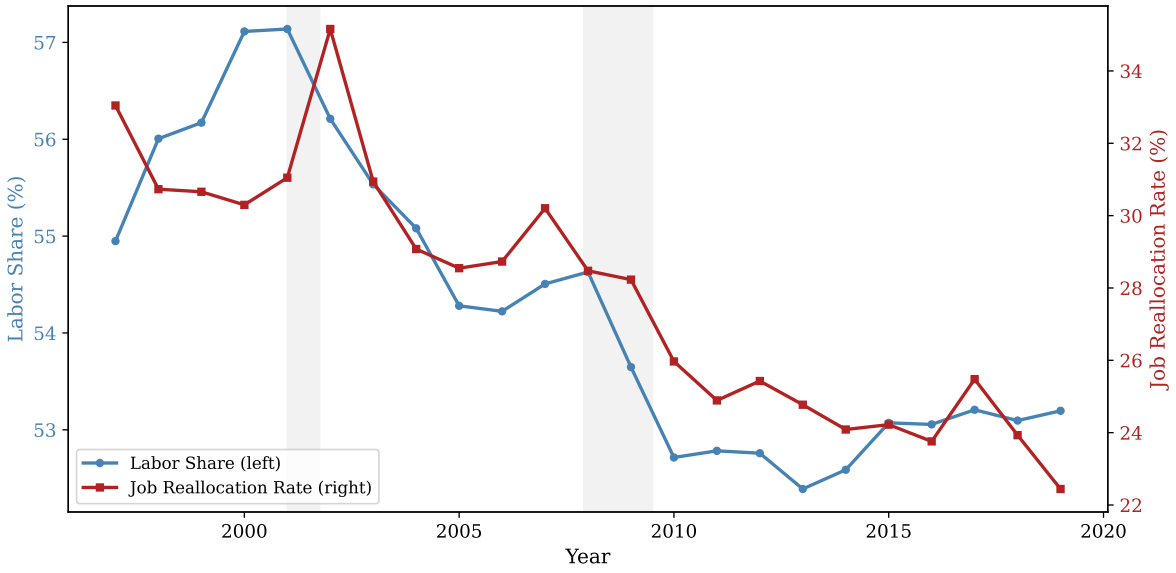


Figure 1: U.S. Labor Share and Job Reallocation Rate, 1997–2019

Notes: This figure plots the national labor share (left axis) and job reallocation rate (right axis). The labor share is compensation of employees divided by GDP, from the BEA Regional Economic Accounts. The job reallocation rate is the sum of national job creation and destruction rates from the Census Business Dynamics Statistics. Shaded regions indicate NBER recession dates.

provides a coarse measure of employment concentration, with limited information about markups. Panel C reports the distribution of long-differences: the average state-industry cell experienced a 4.8 percentage-point decline in the payroll share and an 8.9 percentage-point decline in the JRR.

Figure 1 plots the national labor share and job reallocation rate. Both decline over 2001–2019, although their year-to-year paths differ. The decline in job reallocation is pervasive across all 11 industry groups, with no exceptions.[5]

3. Empirical Analysis

The aggregate co-movement between the payroll share and the JRR documented in Figure 1 suggests a systematic link between declining business dynamism and the falling labor share. We now examine this relationship using panel fixed-effects regressions that exploit *within*-cell variation over time, controlling for permanent cross-cell differences and common aggregate shocks.

[5]Appendix Figure B.1 plots the JRR for each industry with fitted trend lines. Appendix Figure B.2 plots each industry's 2001–2019 change in the payroll share against its change in the JRR and shows that industries with larger reallocation declines also experienced larger payroll-share declines.

Our empirical strategy proceeds in four steps. First, we estimate the core JRR–labor-share relationship using panel fixed effects, progressively adding controls that correspond to the leading alternative explanations for the labor share decline (Section 3.1). Second, we examine heterogeneity by industry type, initial concentration, and time period, using sub-period and rolling-window estimates to study the time dimension (Section 3.2). Third, we explore the mechanism through alternative specifications—excess JRR, lagged JRR, creation-destruction decomposition, and margin decompositions (Section 3.3). Finally, we summarize the panel evidence (Section 3.4).

3.1. Panel Analysis

Our primary specification is a panel fixed-effects model using all annual observations:

$$\log \lambda_{ist} = \alpha_{is} + \gamma_t + \beta \log JRR_{ist} + \mathbf{X}'_{ist}\delta + \varepsilon_{ist} \tag{4}$$

where α_{is} are state-industry fixed effects that absorb all permanent differences across cells (industry composition, geography, local regulations), γ_t are year fixed effects that absorb common macro shocks (business cycles, policy changes, aggregate technology trends), and $\mathbf{X}_{ist}$ is a vector of time-varying controls. The coefficient β is thus identified solely from *within-cell* variation over time: does a given state-industry cell's labor share fall in years when its reallocation rate declines, beyond what can be explained by aggregate trends and the leading competing channels?

Table 1 reports the results for our baseline sample (2001–2019). We build up the specification progressively, adding controls that correspond to the leading alternative explanations for the labor share decline. The goal is to determine whether the JRR–labor-share relationship survives once these competing channels are accounted for.

The baseline specification (column 1) yields an elasticity of 0.044: within a given state-industry cell, a 10% decline in the JRR is associated with a 0.4% decline in the payroll share. The most prominent alternative explanation for the labor share decline centers on rising product market power. Autor et al. (2020) argue that technological change and globalization have increasingly favored a small number of highly productive "superstar" firms that capture disproportionate market shares. Because these dominant firms tend to be less labor-intensive, the reallocation of economic activity toward them mechanically reduces the aggregate labor share. Relatedly, De Loecker et al. (2020) document a broad rise in firm-level markups since the 1980s, implying that firms increasingly price

Table 1: Panel Regression Results

	log(Payroll share)					
	(1)	(2)	(3)	(4)	(5)	(6)
Job reallocation rate	0.044***	0.045***		0.045***	0.045***	0.045***
	(0.008)	(0.008)		(0.008)	(0.008)	(0.008)
HHI (concentration)		0.012	0.008	0.013	0.011	0.012
		(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
Union membership				0.003*		0.003*
				(0.001)		(0.001)
Mfg. share					-0.324	-0.305
					(0.250)	(0.249)
Observations	10631	10631	10631	10631	10631	10631

Notes: The dependent variable is the log payroll share (compensation of employees divided by gross value added). The payroll share, the JRR, and the HHI enter in logs. The union membership rate and the manufacturing employment share enter in levels. The sample covers 2001–2019. Column (1) includes only the JRR. Column (2) adds the employment HHI. Column (3) includes only HHI. Column (4) adds the state-level private-sector union membership rate. Column (5) adds the state-level manufacturing employment share, and column (6) includes all controls simultaneously. All specifications include state×industry and year fixed effects. Standard errors are clustered at the state×industry level and reported in parentheses. Union membership and the manufacturing share vary only at the state level. Clustering column (6) at the state level instead gives a job-reallocation coefficient of 0.045 with a standard error of 0.007. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

above marginal cost and retain a larger share of revenue as profits at the expense of both labor and capital. If this same rise in concentration simultaneously reduces competitive entry and exit—thereby lowering measured job reallocation—then our JRR coefficient could reflect the influence of market power on both variables. To test this, we add log employment HHI as a grouped-data proxy for local market concentration (column 2). The JRR coefficient is unchanged at 0.045, and HHI itself enters insignificantly. When we exclude JRR and retain HHI alone (column 3), concentration cannot explain within-cell labor share variation on its own. We therefore conclude that the JRR–labor-share relationship is not explained away by our concentration proxy. Employment HHI at the state-industry level is a coarse proxy, so the test addresses the role of measured concentration. A broader assessment of market power would require direct information on markups. This finding is consistent with Albrecht and Decker (2026), who show that there is no systematic negative correlation between rising markups and declining business dynamism across industries—the sectors with the largest markup increases are not those with the largest dynamism declines.

A second alternative emphasizes declining worker bargaining power. Stansbury and Summers

(2020) argue that the erosion of unions and labor-market institutions has weakened workers' ability to claim rents, contributing to the labor share decline independently of changes in technology or market structure. Declining unionization could also affect our estimates if union strength is related to reallocation as well as to the labor share. Adding state-level private-sector union membership rates (column 4) has no effect on the JRR coefficient, indicating that the association is not absorbed by the measured unionization control.

A third concern is that changing industrial composition could affect both labor shares and reallocation. Elsby et al. (2013) find that the more recent payroll-share decline mainly reflects within-industry changes, although composition mattered historically. Although our state-industry fixed effects absorb permanent between-industry composition differences, deindustrialization could still confound our estimates through state-level equilibrium channels: states losing manufacturing may experience wage spillovers, demand shifts, or labor-market tightening that affect both the JRR and the labor share within non-manufacturing industries. Controlling for state-level manufacturing employment share (column 5) leaves the JRR coefficient unchanged, indicating that the association is not absorbed by this measured manufacturing-share control.

Composition arithmetic reinforces this conclusion: the manufacturing decline moves in the wrong direction to explain the joint pattern at the heart of this paper. Manufacturing is a low-reallocation sector, with job creation and destruction rates persistently below those of services, construction, and retail trade (Davis et al., 1998). A shrinking manufacturing employment share therefore shifts activity toward high-churn service industries and, holding within-sector reallocation rates fixed, would *raise* the aggregate JRR. Structural transformation away from manufacturing may thus be consistent with a falling labor share, but it cannot account for the contemporaneous decline in job reallocation.

Finally, column 6 includes all three controls simultaneously alongside JRR. The JRR coefficient remains 0.045 and significant at the 1% level. Neither concentration nor manufacturing exposure enters significantly. Union membership carries a small positive coefficient (0.003, $p < 0.10$), the sign the bargaining-power account predicts: a one-point higher union membership rate is associated with a 0.3 percent higher payroll share. The main takeaway from Table 1 is therefore straightforward: the JRR–labor-share relationship is not absorbed by the leading alternative mechanisms we can

Table 2: Heterogeneity Analysis: Panel Regressions

	By Industry		By Concentration		By Period	
	Goods	Services	Low HHI	High HHI	2001–2007	2008–2019
log(JRR)	0.062	0.045***	0.072***	0.033***	0.032***	0.017*
	(0.042)	(0.007)	(0.012)	(0.009)	(0.011)	(0.009)
log(HHI)	0.053*	-0.015	-0.053***	0.029	-0.023	-0.009
	(0.030)	(0.014)	(0.016)	(0.021)	(0.015)	(0.016)
Observations	1927	8704	5244	5212	3904	6727

Notes: The dependent variable is the log payroll share, and all regressors enter in logs. Columns (1)–(2) split the sample into goods-producing and service industries. Columns (3)–(4) split by below- and above-median initial employment HHI, and columns (5)–(6) compare 2001–2007 with the Great Recession and afterward (2008–2019). All specifications include state×industry and year fixed effects and control for log HHI. Standard errors are clustered at the state×industry level and reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

measure in this setting.[6]

3.2. Heterogeneity

The baseline estimates show a positive relationship between job reallocation and the labor share after controlling for concentration, unionization, and manufacturing exposure. We next examine where this association is strongest and whether it persists across different time periods. Table 2 reports separate panel regressions by industry type, initial concentration, and time period.

By industry type. The JRR–labor-share relationship is precisely estimated and highly significant in service industries, which account for the vast majority of U.S. employment. The goods-producing coefficient is larger in magnitude (0.062) but statistically insignificant, reflecting the limited cross-sectional variation available from only two broad goods-producing industry groups (construction and manufacturing). The key finding is that the relationship is most precisely estimated in services—the part of the economy where the labor share decline has the greatest aggregate consequence.

By initial concentration. In lower-concentration cells (below-median HHI), the JRR coefficient is large (0.072) and highly significant. In higher-concentration cells, it is smaller (0.033) but still

[6]To examine the longer-run relationship, we regress the 2001–2019 change in log payroll share on the change in log JRR, controlling for changes in log HHI, union membership, and the manufacturing employment share. Appendix Table B.1 shows that cells with larger reallocation declines also experienced larger labor-share declines over the full period. The annual fixed-effects coefficient can also reflect local cyclical variation, whereas the long difference compares full-period changes across cells.

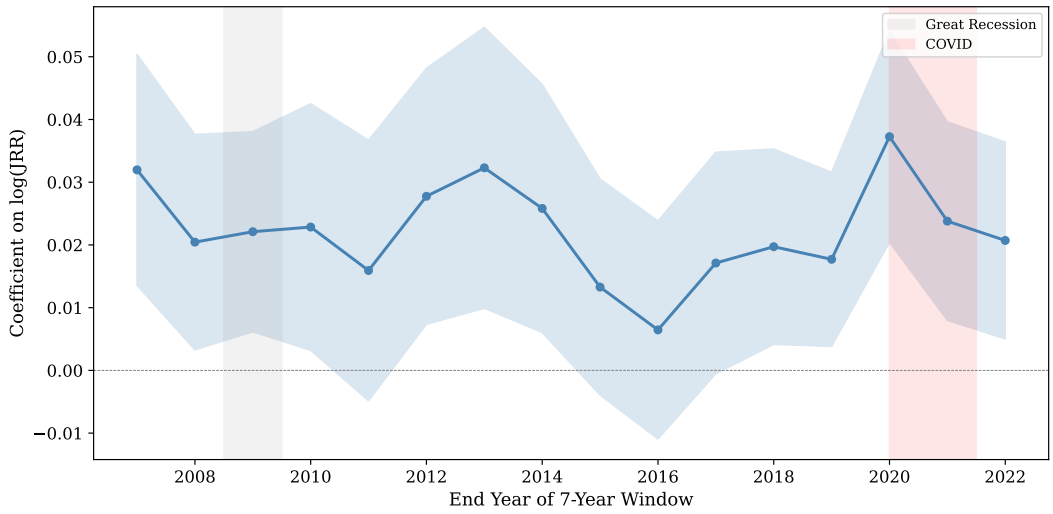


Figure 2: JRR–Labor Share Relationship over Rolling Seven-Year Windows

Notes: Each point is the coefficient on log(JRR) from a panel regression of log(payroll share) on log(JRR), controlling for log employment HHI, with state-industry and year fixed effects, estimated over a 7-year window ending at the indicated year. Shaded bands are 90% confidence intervals (standard errors clustered at the state×industry level). Gray and red shaded regions mark the Great Recession and COVID periods, respectively. Job reallocation from Census BDS.

significant. This gradient is consistent with a labor-friction channel operating more strongly where the grouped-data concentration proxy is least likely to dominate the interpretation. Put differently, the JRR–labor-share relationship is larger in lower-concentration environments, though we do not interpret this proxy as a definitive measure of product-market power.

By time period. The point estimate is larger before the Great Recession (2001–2007, coefficient 0.032, $p < 0.01$) than during the Great Recession and afterward (2008–2019, coefficient 0.017, $p < 0.10$), with a positive relationship in both sub-periods (columns 5–6 of Table 2). Figure 2 follows the relationship across rolling seven-year windows, extending through 2022. The coefficients remain positive in every window. They are smallest in the windows covering the recovery years 2009–2016, where the 90% interval includes zero, and recover in the windows ending in 2018 and 2019. The positive association therefore persists across the sample, with variation in its magnitude.[7]

3.3. Robustness and Alternative Specifications

We next examine alternative specifications and the components of job reallocation.

[7]Appendix Table B.2 reports additional sub-period estimates. Excluding 2008–2009 gives a coefficient of 0.017 for 2010–2019. Excluding the immediate recovery years, 2010–2011, gives 0.024 ($p < 0.01$) for 2012–2019.

Table 3: Robustness and Alternative Specifications

Specification	Coefficient	(s.e.)	N
Excess JRR	0.031***	(0.006)	10631
Lagged JRR ($t-1$)	0.030***	(0.008)	10070
JC rate (entered jointly)	0.012*	(0.006)	10631
JD rate (entered jointly)	0.033***	(0.005)	
Intensive margin (continuers)	0.019**	(0.007)	10631
Extensive margin (births+deaths)	0.027***	(0.004)	10631

Notes: The dependent variable is the log payroll share. All specifications use 2001–2019 BDS data with state×industry and year fixed effects. The "excess JRR" is job reallocation in excess of the absolute net employment change, $JRR - |JCR - JDR|$, the excess reallocation measure of Davis et al. (1998). The "entered jointly" rows report coefficients from a single regression with log JC and log JD entered simultaneously. The intensive margin is job reallocation from expansions and contractions of continuing establishments. The extensive margin is job creation from establishment births plus job destruction from establishment deaths. Standard errors are clustered at the state×industry level and reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 3 reports the key checks. The *excess JRR* specification uses job reallocation in excess of the absolute net employment change, $JRR - |JCR - JDR|$, the excess reallocation measure of Davis et al. (1998). It yields a significant coefficient of 0.031. The association is therefore present in the simultaneous creation and destruction of jobs within a cell and does not rest on reallocation that accompanies net expansion or contraction.

The *lagged JRR* specification (0.030, $p < 0.01$) checks whether the relationship extends beyond contemporaneous co-movement. Its positive coefficient shows that prior-year reallocation is associated with the current payroll share, consistent with the conditional-association interpretation of the panel evidence.

When job creation and destruction enter jointly, the job destruction coefficient is larger and more precisely estimated (0.033, $p < 0.01$) than the job creation coefficient (0.012, $p < 0.10$). We interpret these coefficients as differences in the strength of the conditional associations.

The *entry/exit decomposition* is also informative. The BDS reports job flows from establishment births and deaths separately from continuing establishments, allowing us to decompose the JRR into an entry/exit component (births + deaths) and a continuing-establishment component (expansions + contractions of surviving establishments). Both components are significant when entered separately: the entry/exit margin (0.027, $p < 0.01$) and the continuing-establishment margin (0.019, $p < 0.05$). That both contribute is natural, since multi-unit firms can reallocate employment across their establishments and open or close units. While the entry/exit estimate is larger and more precise, we

read the comparison as a difference in the strength of the conditional associations rather than as a ranking of channels.

Finally, we verify that our results are not driven by the inclusion of employer-provided supplements (such as health insurance) in the compensation numerator. Replacing the compensation share with a *wages-only share*—wages and salaries divided by GDP, excluding employer supplements—leaves the panel coefficient essentially unchanged (Appendix Table B.3). The relationship therefore also holds for wage payments alone.

3.4. Discussion

Taken together, the panel results document a within-cell association between job reallocation and the payroll share that persists after adding the measured controls and varies across markets, industries, and margins.

First, it survives stringent fixed effects and controls. State-industry and year FE absorb permanent cross-cell differences and common aggregate shocks. The JRR coefficient is not absorbed by adding HHI, unionization, and manufacturing controls—the primary confounders implied by the superstar-firm, declining-bargaining-power, and deindustrialization hypotheses. Second, the relationship is larger in lower-concentration markets than in higher-concentration markets. Third, the relationship is most clearly estimated in services. Fourth, the estimated association is larger for job destruction than for job creation. Fifth, it is larger for entry and exit than for continuing establishments, though both contribute. Replacing total compensation with wages only leaves the panel coefficient unchanged, showing that the association is also present for wage payments alone.

These panel estimates establish conditional associations without identifying the source of declining reallocation. We next use public BED tabulations to examine the distribution of quarterly establishment employment changes and connect the inaction margin to the state-industry panel. Rising adjustment costs offer one explanation for these patterns, which the structural model in Section 5 evaluates quantitatively.

4. The Rise of Inaction and Employment Adjustment

The panel evidence links reallocation to payroll shares. We now examine the establishment behavior underlying declining reallocation using the BED size-of-change data: how often

establishments change employment and how large those changes are. These tabulations complement the annual BDS flows and allow us to connect establishment inaction to the state-industry panel.[8]

4.1. The Rise of Inaction

The aggregate JRR can decline through two channels: establishments may adjust less *frequently*—a rising share of units reporting zero employment change—or they may adjust by smaller amounts when they do adjust. We refer to the first as the *inaction margin* and the second as the *size-of-change margin*. Note that these are conceptually distinct from the entry/exit decomposition in Section 3.3, which partitions the JRR by establishment demographic status (births and deaths versus continuing establishments). Appendix Table B.5 decomposes the decline in the constructed gross-flow rate using the employment exposure described there. Falling adjustment incidence accounts for about half of its log decline. Lower conditional intensity is the largest contributor, partly offset by the rising relative size of adjusting establishments. The structural model asks whether higher adjustment costs can generate the observed rise in inaction and shift toward smaller adjustments.

The inaction margin: more establishments with unchanged employment. Panel (a) of Figure 3 plots the inaction rate—the fraction of private-sector establishments reporting zero net employment change—against the aggregate job reallocation rate, showing a strong mirror-image pattern. The inaction rate was approximately 44% in the late 1990s and has risen steadily, reaching 52% by 2019.[9] Inaction rises while reallocation and the labor share decline over 2001–2019, although their paths differ. As the fraction of establishments with unchanged employment rises, the JRR falls.[10]

[8]The two sources tabulate different things: the BED classifies establishments by the size of their quarterly employment change, whereas the BDS reports annual job flows split by establishment births, deaths, and continuers. Both build job flows from establishment-level employment changes. They differ in coverage (QCEW versus the Longitudinal Business Database) and timing (quarterly versus March-to-March). Using annual averages of quarterly BED reallocation also yields a positive panel association (Appendix Table B.4). This regressor differs in horizon from March-to-March BDS reallocation.

[9]The inaction rate in Figure 3 is establishment-count based: each establishment counts equally, regardless of size. An employment-weighted variant divides employment at no-change establishments by average-endpoint employment reconstructed from BED job-flow levels and rates. It rises from 14.3% in 2001 to 16.6% in 2019. Its lower level reflects the smaller average size of no-change establishments.

[10]The inaction series carries two measurement caveats. First, inaction requires exact equality of reported employment in the third month of consecutive quarters. Reporting errors can therefore create or conceal a zero change. We use the published BED series, which incorporate BLS corrections and revisions, but the available tabulations do not quantify remaining misclassification. Second, quarterly snapshots measure net employment changes. Offsetting hires and separations, or changes reversed between observation dates, can leave employment unchanged. We apply the same quarterly net-change definition to the model and compare the trend at an annual frequency. The March-to-March

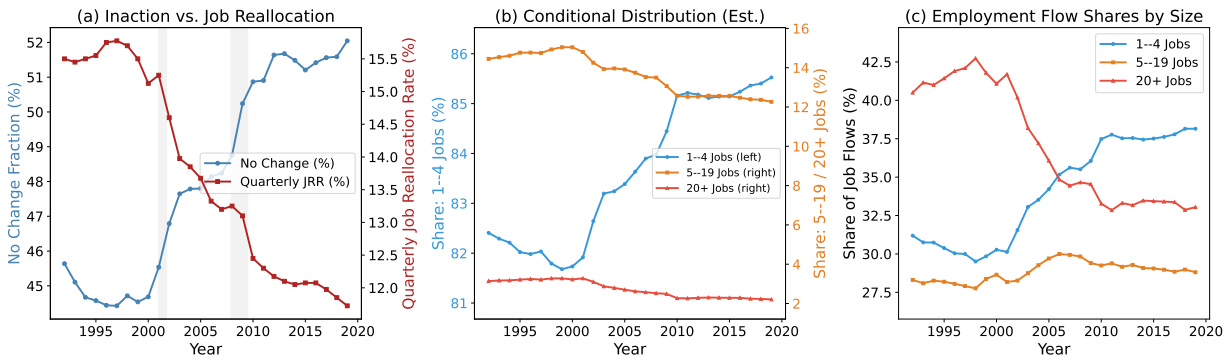


Figure 3: The Rise of Inaction and the Shift in the Size-of-Change Distribution, 1992–2019

Notes: The inaction rate gives each establishment equal weight and measures zero net employment change. Panel (a) overlays the quarterly inaction rate—the fraction of establishments reporting zero net employment change between consecutive quarters (left axis)—with the national quarterly gross job reallocation rate (right axis), the sum of BED gross job gains and gross job losses rates, including openings and closings. Panel (b) shows the conditional distribution of changing establishments by size of change. The small-adjustment share (1–4 jobs, left axis) has risen while the medium (5–19) and large (20+) shares (right axis) have declined. Panel (c) shows the share of total employment flows (job gains plus losses) accounted for by each size category. All series are calendar-year averages of quarterly observations. Gross job reallocation rates are seasonally adjusted, while inaction and size-of-change measures are not. Shaded areas indicate NBER recession dates. *Source*: Bureau of Labor Statistics, Business Employment Dynamics.

The size-of-change margin: smaller adjustments among adjusters. Panels (b) and (c) of Figure 3 examine the conditional distributions among establishments that *do* adjust. Panel (b) shows the distribution of changing establishments by size of change: among establishments that adjust, the share making small adjustments (1–4 jobs) has risen from 82% to 86% (left axis), while the share making medium (5–19 jobs) and large adjustments (20 or more jobs) has fallen (right axis). Panel (c) shows the corresponding employment flow shares: the fraction of total job flows accounted for by small-change establishments (1–4 jobs) has risen from approximately 30% to 38%, while the share from large-change establishments has declined.

These count-based and employment-based measures answer different questions. The no-change rate records how widespread inaction is across establishments. Gross job flows also depend on the number of jobs adjusted by each adjusting establishment, relative to aggregate employment. Inaction at a small establishment contributes as much to the count-based rate as inaction at a large one, but removes fewer potential job flows. A rise in the no-change rate therefore does not by itself determine the decline in reallocation. The size of the adjustments that no longer take place matters

no-change rate also rises, from 29.8% in 2001 to 36.4% in 2019, showing that the increase is not specific to quarterly measurement. Further details appear in Appendix A.3.

as well. We return to this distinction when choosing the counterfactual targets in Section 6.

4.2. Bridging Establishment Inaction to the BDS Panel

The national BED size-of-change data in Figure 3 establish the aggregate rise of establishment-level inaction, but our main panel regressions are estimated on state-industry cells using annual BDS job flows. To connect these two pieces of evidence more directly, we construct a state-industry inaction proxy from the BED's published quarterly rates of establishments gaining and losing employment by state and sector.[11]

Table 4 relates the BED inaction proxy to BDS job reallocation and payroll shares within state-industry cells, in panel fixed-effects regressions and in long differences. State-industry cells with larger increases in establishment-level inaction experienced larger declines in BDS job reallocation.[12] The two specifications use different variation. The panel regressions ask whether years with higher inaction within a state-industry cell also have lower reallocation or payroll shares, after accounting for state-industry and year fixed effects. The long differences compare changes between the initial and final three-year averages, emphasizing the persistent component of the relationship. Together, they connect the national trends to variation across local industry labor markets. The negative payroll-share coefficient also shows that the bridge extends beyond a relationship between two measures of employment adjustment.

The bridge results sharpen the interpretation of the main panel evidence in three ways. First, the BED inaction proxy is strongly negatively associated with the BDS JRR within the same state-industry cells: in panel FE regressions, a 10% increase in establishment inaction is associated with approximately a 4.7% decline in total reallocation. Second, this relationship is present for both continuing-establishment and birth/death margins, but it is quantitatively larger for the extensive margin. This pattern is important for interpretation: establishment no-change in BED does not map one-for-one into the continuing-establishment component of BDS job flows. Rather, the two datasets appear to capture different parts of a broader slowdown in employment adjustment, one

[11] Specifically, let g, ℓ, o, and c denote the BED gaining, losing, opening, and closing rates in percent, and put $b = 100 - g - \ell$ and $t = (o+c)/2$. The fraction of establishments with no employment change is $q_U = 100(b+t)/(100+t)$. We compute it quarterly and then average within each year. This converts the BED average-endpoint count denominator to the union of establishments present at either endpoint. At the national level, the rate-implied measure agrees with the direct size-of-change counts within published rounding over 2001–2019.

[12] Appendix Figure B.3 shows this relationship visually.

Table 4: Bridging BED Inaction to BDS Reallocation and Payroll Shares

	(1)	(2)	(3)	(4)	(5)
Panel A: State-industry panel fixed effects, 2001–2019					
Dependent variable	log JRR	log intensive JRR	log extensive JRR	log payroll share	log payroll share
log(BED inaction)	-0.474***	-0.326***	-0.802***	-0.175***	-0.152***
	(0.062)	(0.055)	(0.107)	(0.050)	(0.050)
log(BDS JRR)					0.049***
					(0.007)
N	9473	9473	9473	9473	9473
R^2 (within)	0.122	0.078	0.121	0.063	0.107
Panel B: State-industry long differences, 2001–2003 to 2017–2019					
Dependent variable	Δ log JRR	Δ log intensive JRR	Δ log extensive JRR	Δ log payroll share	Δ log payroll share
Δ log(BED inaction)	-0.367***	-0.147*	-0.833***	-0.305***	-0.240***
	(0.085)	(0.077)	(0.159)	(0.075)	(0.079)
Δ log(BDS JRR)					0.176***
					(0.051)
N	499	499	499	499	499
R^2	0.048	0.009	0.064	0.030	0.056

Notes: BED inaction is the unchanged-establishment share of the union of establishments active at either quarterly endpoint, reconstructed from published gaining, losing, opening, and closing rates and then averaged annually (Appendix A.3). Panel A uses state-industry and year fixed effects with standard errors clustered by state-industry cell. Panel B uses state-industry long differences between 2001–2003 and 2017–2019, with HC1 robust standard errors. The intensive JRR is the continuing-establishment margin from BDS. The extensive JRR is the births-plus-deaths margin. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

that affects both adjustments at continuing establishments and establishment turnover. Third, the bridge extends to the labor share itself. Cells with larger inaction increases also experienced larger payroll-share declines, and in a horse-race specification both the BDS JRR and BED inaction remain significant. The long-difference results point in the same direction, and the extensive-margin and payroll-share bridges remain strong.[13] The BED and BDS evidence thus provide complementary information: the BED inaction series disciplines the micro margin of adjustment most directly, while the BDS panel shows where broader reallocation declines coincide with lower labor shares.

4.3. Implications for Modeling

This section has documented two margins of declining reallocation, fewer establishments adjusting and smaller adjustments among those that do, and has linked the first to lower reallocation

[13]The bridge varies across the sample. Appendix Table B.6 shows a larger continuing-establishment coefficient in magnitude in lower-concentration cells, and a stronger bridge from inaction to payroll shares before the Great Recession.

and payroll shares across state-industry cells. Whether rising adjustment costs can account for the aggregate declines, and by how much they lower the labor share, depends on equilibrium responses that the data cannot reveal: how costlier adjustment changes the value of an establishment, the wage at which entry breaks even, and the allocation of employment. Section 5 builds a model with these features, and the evidence here guides two of its choices: the shape of the adjustment cost and the scope of the model. Rising inaction and smaller adjustments jointly inform the shape of the cost. A kinked per-worker cost creates a gap between the marginal returns required to expand and contract, supporting an inaction region. A quadratic component makes the marginal cost increase with adjustment size and can shift changes toward smaller bins. Because more than one cost shape can generate inaction, the no-change rate alone does not identify which one is at work.

The size distribution provides additional discipline. A fixed cost per adjustment event encourages larger, less frequent changes, whereas a cost that rises with the number of jobs adjusted can limit their size. We therefore allow fixed, kinked, and quadratic components and use the joint changes in inaction and gross flows to discipline the counterfactual increase in costs. The conditional adjustment-size distribution then provides an untargeted counterfactual check (Section 6.2).

The bridge results also bear on the scope of the model. The link between inaction and reallocation appears on both the continuing-establishment and the birth/death margins. An establishment's decision to leave employment unchanged is distinct from its birth or death, but a change in the cost of operating and adjusting can affect both incumbents' policies and the value of operating an establishment. We therefore model entry and exit endogenously and measure total reallocation in the model from simulated establishment histories, including openings and closings, which is the object the annual BDS measure records.

The empirical-to-model mapping thus has three steps. The state-industry estimates establish the joint movement of inaction, reallocation, and payroll shares. The national BED moments discipline the employment-adjustment decisions that can generate this movement. Finally, the model solves for the equilibrium wage and labor share that these policy changes imply. We calibrate to the levels and changes of these national moments. The counterfactual then incorporates the wage and establishment-distribution responses that link individual adjustment to aggregate outcomes.

4.4. Evidence on Rising Adjustment Costs

Other evidence is consistent with the adjustment-friction interpretation. Decker et al. (2020) find that employment has become less responsive to productivity since 2000, and Cooper et al. (2024) examine rising labor adjustment costs directly.

Three kinds of costs could have risen. First, regulatory compliance: Trebbi et al. (2026) estimate that compliance absorbs 1.3–3.3% of the average firm's wage bill and document rising compliance costs over 2002–2014. Second, recruitment and adaptation costs: Blatter et al. (2012) and Muehlemann and Pfeifer (2016) show that measured hiring costs amount to several weeks of wages per hire, including recruitment, training, and initial productivity losses.[14] Advance-notice requirements for large layoffs are a cost that rises with the size of the change (Addison and Blackburn, 1994), and noncompete agreements and occupational licensing also constrain worker mobility (Starr et al., 2021; Kleiner and Krueger, 2013). Third, the administration of employer-provided benefits at hiring and separation. Adding a worker requires enrolling the worker in health and other benefit plans, and a separation requires terminating coverage and issuing the continuation-coverage notices that federal law requires. These tasks are per-worker costs of changing headcount, and the benefit package they administer has grown: in the Kaiser Family Foundation Employer Health Benefits Survey, the average annual employer contribution to family health insurance premiums approximately doubles between 2001 and 2019 (Appendix Figure B.4).[15] The quantitative exercises in Sections 5 and 6 discipline the effective cost increase using establishment adjustment behavior.

5. Model

We documented rising inaction and smaller employment adjustments, linked rising inaction and declining reallocation to lower payroll shares in state-industry data, and set out the choices a model of this behavior must make. This section builds that model. It is a general equilibrium model of heterogeneous firms in the tradition of Hopenhayn and Rogerson (1993). Employment adjustment costs combine fixed, kinked per-worker, and quadratic components. The model is calibrated to the frequency and measurement conventions of the BED data, including count-based and employment-

[14] These studies measure the level and shape of hiring costs, rather than their change over our sample.

[15] Appendix B.7 provides further details on the evidence for rising adjustment costs.

weighted quarterly inaction and three points of the conditional gains-size distribution. We use it to quantify the implications of rising adjustment costs for annual reallocation and the labor share.

5.1. Environment

The economy comprises a continuum of heterogeneous firms and a representative household. Firms use labor as their sole input to produce a homogeneous good. Firm-level dynamics are governed by persistent and transitory idiosyncratic shocks and endogenous entry and exit decisions.

Incumbent Firms. An incumbent firm begins period t with state (s_{t-1}, n_{t-1}), where s_{t-1} is its previous-period persistent productivity and n_{t-1} is its previous employment. Its value is $W(s_{t-1}, n_{t-1})$.

Exit/Stay Decision. The firm first draws a stochastic exit value x_t from distribution $\Psi(\cdot)$ and then decides whether to exit. If it exits, it collects x_t but must pay a per-worker dismissal cost γn_{t-1} to release its workforce. If it stays, it observes its new persistent productivity s_t and a transitory shock ε_t and proceeds to the adjustment decision. The integrated value is:

$$W(s_{t-1}, n_{t-1}) = \int \max\left(x_t - \gamma n_{t-1},\, \mathbb{E}\left[V^c(s_t, \varepsilon_t, n_{t-1}) \,\middle|\, s_{t-1}\right]\right) d\Psi(x_t)$$

where the expectation is taken over both the persistent transition $s_t \mid s_{t-1}$ and the i.i.d. transitory draw ε_t.

Adjust/Inaction Decision. A firm that stays, now knowing its state $(s_t, \varepsilon_t, n_{t-1})$, must decide whether to adjust its labor force or do nothing:

$$V^c(s_t, \varepsilon_t, n_{t-1}) = \max\left\langle \underbrace{V^a(s_t, \varepsilon_t, n_{t-1})}_{\text{Adjust}}, \underbrace{V^n(s_t, \varepsilon_t, n_{t-1})}_{\text{Do Nothing}} \right\rangle$$

This explicit choice is the core mechanism. If the shock is not large enough to warrant paying the adjustment cost, the firm optimally chooses inaction (V^n), setting its net employment change to zero. This maps directly to the rising mass of "no-change" firms in Figure 3.

If the firm does not adjust:

$$V^n(s_t, \varepsilon_t, n_{t-1}) = f(s_t, \varepsilon_t, n_{t-1}) - w_t n_{t-1} + \beta\, W(s_t, n_{t-1})$$

If the firm adjusts to a new labor level n_t:

$$V^a(s_t, \varepsilon_t, n_{t-1}) = \max_{n_t} \{f(s_t, \varepsilon_t, n_t) - w_t n_t - h(n_t, n_{t-1}) + \beta W(s_t, n_t)\}$$

where the continuation value W already integrates over the exit draw (as defined above) and, because the transitory shock is i.i.d., depends only on the persistent state (s_t, n_t). The expectation over next-period shocks is embedded recursively through the fixed-point structure of W.

Employment is a state variable because today's workforce becomes the starting point for tomorrow's adjustment. The benefit of changing employment therefore includes both the current profit gain and the change in continuation value. A persistent productivity increase raises the return to expansion over several quarters, whereas a transitory increase raises current revenue without improving the future productivity state. Paying to expand in response to the latter can also create a subsequent need to contract. Firms weigh these current and future returns against the cost of moving away from inherited employment. Inaction is optimal when no feasible change produces a sufficient gain in the combined value.

The adjustment cost combines a fixed component, a kinked per-worker component, and a quadratic component:

$$h(n_t, n_{t-1}) = F\,\mathbb{1}\{n_t \neq n_{t-1}\} + \gamma|n_t - n_{t-1}| + \tfrac{1}{2}\phi(n_t - n_{t-1})^2. \tag{5}$$

The parameter F is paid whenever employment changes, γ is a per-worker cost paid on each job added or destroyed, and ϕ governs the quadratic cost. The fixed term makes small changes less attractive, while the kink creates a gap between the marginal returns required to expand and contract. The quadratic term makes marginal cost increase with adjustment size. Kinked per-worker hiring and firing costs are standard in dynamic labor-demand models (Bentolila and Bertola, 1990; Hopenhayn and Rogerson, 1993; Cooper et al., 2007). With integer employment, all three terms affect the cost of a one-worker change, so we discipline the parameters using several adjustment moments jointly.

For a change of $k \geq 1$ whole jobs, the cost is $F + \gamma k + \phi k^2/2$, and increasing its size from k to $k+1$ costs $\gamma + \phi(k + 1/2)$. The per-worker component adds the same amount to each job adjusted. The quadratic component adds progressively more as the change grows. A one-job change costs

$F + \gamma + \phi/2$, so all three parameters influence the decision to make the smallest adjustment.[16] Larger changes reveal the additional role of curvature. Observing both the frequency and size of employment changes is consequently useful for disciplining the cost schedule.

Potential Entrants. In any period, a mass of potential entrants can enter by paying a sunk entry cost c_e. Upon entry, a firm draws its initial persistent productivity from distribution $\nu(s_t)$ and observes a transitory shock. It chooses its initial workforce without paying labor adjustment costs. Its first-period value is

$$V^{e,0}(s_t, \varepsilon_t) = \max_{n_t} \{ f(s_t, \varepsilon_t, n_t) - w_t n_t + \beta W(s_t, n_t) \}.$$

The chosen workforce becomes inherited employment in the next period, when the incumbent adjustment cost applies. The value of entry is

$$V^e = \int V^{e,0}(s_t, \varepsilon_t)\, d\nu(s_t)\, dG(\varepsilon_t).$$

Free entry ensures $V^e = c_e$ whenever entry is positive. This condition pins down the equilibrium wage, and the mass of entrants is then set by labor-market clearing.

Households. A representative household owns all firms, supplies labor, and consumes. As in Lee and Mukoyama (2018), we assume utility is linear in consumption ($u(C_t) = C_t$), so firms discount future profits by β. The household's labor supply is determined by:

$$v'(L_t) = w_t$$

where $v(\cdot)$ is the disutility of labor. We never need to specify $v(\cdot)$: as explained below, the labor share and every rate we report are invariant to the mass of establishments, which is the only equilibrium object the labor-supply condition determines.

[16] With continuous employment and a smooth interior optimum, quadratic costs alone do not create an inaction region. On the integer portion of the employment grid, the smallest change is one worker and costs $\phi/2$ even when $F = \gamma = 0$, so a small shock can leave employment unchanged.

5.2. Stationary Equilibrium

A stationary equilibrium is characterized by a wage w^*, a mass of entrants M^*, and a distribution of incumbents $\mu^I(s,n)$ such that firms optimize, free entry holds, the distribution is invariant under employment and exit policies and entrant inflows, and the labor market clears:

$$L^s(w^*) = \int n^I(s,n,\varepsilon;w^*)\, d\mu^I(s,n)\, dG(\varepsilon) + M^* \int n^e(s,\varepsilon;w^*)\, d\nu(s)\, dG(\varepsilon).$$

Here n^I and n^e are the incumbent and entrant employment policies. The free-entry condition alone pins down w^*. Given w^*, these policies and the stationary distribution per unit of entrant mass are determined, and labor-market clearing sets M^*. Because M^* scales aggregate employment and output proportionally, the labor share wL/Y and every rate we report are invariant to it, and hence to the labor-supply curve. We therefore normalize entrant mass to one, report aggregates per unit of entrant mass, and leave $v(\cdot)$ unspecified.[17] Aggregate labor supply determines the scale of the establishment population.

5.3. Calibration

We calibrate the benchmark economy to the U.S. economy in 2001, the initial year of our empirical panel. Ten moments jointly discipline ten parameters, combining aggregate outcomes with the frequency and size of establishment adjustments. We then hold the remaining parameters fixed and raise the kink and quadratic cost parameters to match the 2001–2019 changes in quarterly inaction and gross job flows. Changes in the labor share, annual reallocation, and conditional adjustment sizes are untargeted counterfactual predictions.

Production is $f(s,\varepsilon,n) = s\, e^{\varepsilon}\, n^{\theta}$, where θ governs the labor elasticity of output. In a frictionless economy, θ equals the labor share. The idiosyncratic shock has a persistent and a transitory component. The persistent component follows a log-AR(1), $\log s_t = a + \rho \log s_{t-1} + \eta_t$, whose innovation η_t is drawn from a variance-preserving two-component mixture of normals: with probability p it comes from a "jump" component ξ times wider than the baseline component, which is rescaled so that the total innovation variance remains σ_η^2. The mixture parameters (p,ξ) thus govern the kurtosis of the innovation while leaving its persistence and dispersion unchanged—which

[17] The reported employment changes measure labor demand per unit of entrant mass at the new equilibrium wage.

Table 5: Model Calibration

Parameter	Value	Description
Panel A: Externally set (quarterly)		
β	$0.96^{1/4}$	Discount factor
ρ	0.983	Persistence of persistent productivity
σ_η	0.119	Persistent-innovation volatility
Panel B: Internally calibrated (2001)		
θ	0.612	Labor elasticity of output
c_e	1553	Sunk entry cost
a	0.018	Productivity drift
σ_ε	0.204	Transitory shock volatility
p	0.018	Jump probability
ξ	11.8	Jump-size multiple
x_0	0.843	Additional exit-shock mass at zero
F_0	0.0038	Fixed cost per adjustment event
γ_0	0.014	Per-worker adjustment cost
ϕ_0	0.045	Quadratic adjustment cost

Notes: Panel A gives the parameters externally set at the quarterly values. Panel B parameters are jointly calibrated to the ten 2001 moments in Table 6.

is what lets the model reproduce the fat-tailed establishment growth-rate distribution. The transitory component e^{ε_t} is an i.i.d. mean-one shock, $\varepsilon_t \sim N(-\sigma_\varepsilon^2/2, \sigma_\varepsilon^2)$.[18] The exit value distribution mixes an additional mass x_0 at zero with a uniform distribution on 200 grid points between zero and 2,000.

Table 5 reports the model calibration. The model is calibrated at quarterly frequency with the discount factor $\beta = 0.96^{1/4}$. The persistent process is set externally at $\rho = 0.983$ and $\sigma_\eta = 0.119$, using quarterly conversions of an annual process with persistence 0.934 and innovation standard deviation 0.233, following Cooper and Haltiwanger (2006).[19] Ten parameters (θ, c_e, a, x_0, γ_0, ϕ_0, F_0, σ_ε, p, and ξ) are jointly calibrated to the labor share, annual job reallocation, establishment entry, average establishment size, quarterly count-based and employment-weighted no-change, quarterly

[18] Employment is chosen on an integer grid, augmented with a fine geometric tail so that large establishments can make small proportional adjustments. All discretization and simulation details are in Appendix C.

[19] The raw persistent process implies unconditional dispersion of 0.65. Annual productivity measured on the simulated establishment panel has persistence of 0.89. The simulated measurement includes selection and the transitory shock.

gross job flows, and the conditional 1–4, 20+, and 50+ shares among gaining establishments.[20]

The moments discipline the parameters jointly. Around the benchmark, a smaller transitory shock and a larger quadratic cost both raise inaction and the small-adjustment share and lower quarterly gross flows. They differ in their effects on the upper tail. The 20+ and 50+ shares therefore help separate shocks from costs. Employment-weighted inaction adds information about the size of establishments that leave employment unchanged. The entry cost, the exit-value mass x_0, the drift, and θ also affect adjustment through the equilibrium wage. Appendix Table C.1 reports the local responses to all ten parameters. These responses describe which moments provide information about each parameter without assigning each moment to a single parameter.

The quarterly and annual flow moments also help distinguish persistent from transitory adjustment incentives. Transitory shocks can generate changes that reverse within a year, while persistent shocks give establishments a reason to maintain a changed workforce. An establishment that adds five jobs and later removes five contributes to quarterly gross flows but not to annual reallocation, so the annual measure depends on the persistence and reversal of quarterly decisions. Adjustment costs influence both responses by changing the value of acting immediately.

Table 6 shows that nine of the ten targeted moments lie within 10 percent of their empirical counterparts, including count-based and employment-weighted inaction and the three adjustment-size shares. The shock mixture helps reproduce the concentration of small changes alongside the large-adjustment tail.[21] Quarterly gross flows are lower, at 11.2% against 15.3%. Around the benchmark, increasing transitory volatility raises gross flows while lowering inaction and the conditional small-adjustment share. The joint calibration balances the fit of these related moments.

Panel B reports selected untargeted moments. Annual job creation, job destruction, establishment exit, and inaction closely match their data counterparts. Model entrants average 5.2 employees, compared with 5.6 in the data. These comparisons support using the benchmark to study how higher costs change establishment employment decisions.

[20]The BDS and BED targets connect the calibration to the annual and quarterly evidence in Sections 3 and 4. Each parameter vector is evaluated at the wage satisfying free entry, using a common simulated panel. The loss and measurement conventions are described in Appendix C.6.

[21]With Gaussian persistent innovations and the same cost specification, grid, targets, and weights, recalibrating the remaining eight parameters gives a 1–4 share of 70.2% against 82.4% and a 50+ share of 0.28% against 0.87%. The returned loss is about eight times the benchmark's.

Table 6: Model Fit: Targeted and Selected Untargeted Moments

Moment (%, unless noted)	Model	Data (2001)
Panel A: Targeted moments		
Labor share	57.2	57.1
Job reallocation rate (annual)	31.50	31.05
Establishment entry rate (annual)	10.2	11.0
Average establishment size (employees)	18.9	18.0
Quarterly no-change rate	45.3	45.5
Gross job gains + losses (per quarter)	11.21	15.25
Conditional share: 1–4 jobs	81.3	82.4
Conditional share: 20+ jobs	3.3	3.1
Conditional share: 50+ jobs	0.79	0.87
Employment-weighted no-change	14.3	14.3
Panel B: Selected untargeted moments		
Job creation rate (annual)	15.8	15.6
Job destruction rate (annual)	15.7	15.4
Establishment exit rate (annual)	10.2	10.5
Annual no-change among continuing establishments	38.1	38.7
Average entrant size (employees)	5.2	5.6
Establishment gaining / losing rate (per quarter)	25.8 / 30.4	28.4 / 29.3

Notes: Panel A reports the ten targeted moments. Panel B reports selected untargeted comparisons. Employment-weighted no-change uses employment at no-change establishments divided by average endpoint employment. Annual no-change is conditional on continuing establishments. *Sources:* BEA, Census BDS, and BLS BED.

6. Quantitative Results

In this section, we examine how much rising adjustment costs contribute to the 2001–2019 declines in job reallocation and the labor share using the calibrated general equilibrium model. We compare the 2001 benchmark economy with a stationary equilibrium in which the kink and quadratic cost parameters are higher. We use this pair to allow both the level and the slope of marginal adjustment costs to rise. The kink raises the cost of each job added or cut, while the quadratic term makes marginal costs rise more steeply with adjustment size. The fixed adjustment cost, productivity process, production technology, entry cost, and exit-value distribution stay at their benchmark values. The two cost parameters are chosen to match the observed changes in quarterly inaction and gross job flows. The no-change target restricts how many firms adjust, and the gross-flow target, which weights adjustments by jobs, restricts the total number of jobs added or cut

Table 7: Counterfactual Analysis: Rising Adjustment Costs

	Model		Data	
Moment (%, unless noted)	2001 → CF	Δ (pp)	2001 → 2019	Δ (pp)
Panel A: Targeted changes				
Quarterly no-change rate	45.3 → 51.9	+6.6	45.5 → 52.0	+6.5
Gross job gains + losses (per quarter)	11.21 → 7.65	-3.56	15.25 → 11.73	-3.53
Panel B: Untargeted predictions				
Labor share	57.18 → 56.23	-0.95	57.14 → 53.20	-3.94
Job reallocation rate (annual)	31.50 → 25.62	-5.88	31.05 → 22.44	-8.61
Conditional share: 1–4 jobs	81.3 → 86.2	+4.9	82.4 → 85.5	+3.1
Conditional share: 5–19 jobs	15.4 → 10.5	-4.9	14.5 → 12.3	-2.2
Conditional share: 20–49 jobs	2.50 → 2.99	+0.48	2.25 → 1.66	-0.59
Conditional share: 50+ jobs	0.79 → 0.36	-0.43	0.87 → 0.58	-0.29
Employment-weighted no-change	14.3 → 19.2	+4.9	14.3 → 16.6	+2.3
Establishment entry rate (annual)	10.2 → 10.3	+0.0	11.0 → 9.6	-1.4
Average establishment size (employees)	18.9 → 19.1	+0.3	18.0 → 18.4	+0.4
Equilibrium wage (w^*), log points	0.990 → 0.945	-4.7	—	
Adjustment cost (% of wage bill)	4.1 → 5.0	+0.9	—	
Panel C: Decomposition of the model labor-share change (pp)				
Wage channel	-2.67			
Allocation/productivity channel	+1.72			
Total	-0.95			

Notes: The kink and quadratic cost parameters increase by factors of 1.54 and 2.94, holding the fixed cost and all other parameters constant. Both targets are changes relative to the model benchmark. Adjustment-size shares refer to gaining establishments including openings. The four reported bins are mutually exclusive. Employment-weighted inaction uses average endpoint employment. Average establishment size changes are in employees. Panel C is an exact Shapley decomposition. Wage changes are log points and hold the productivity process fixed. Adjustment cost is continuing-establishment resource expenditure divided by the wage bill. *Sources:* BEA, Census BDS, and BLS BED.

relative to employment. Section 6.2 examines alternative combinations and shows that increasing fixed and quadratic costs gives similar aggregate results.

For each candidate pair of cost parameters, we first solve firms' dynamic employment and exit decisions at a trial wage. We then adjust the wage until the value of entry equals the unchanged sunk entry cost and construct the associated stationary distribution. The simulated quarterly histories yield the no-change and gross-flow moments used to select the cost increase. The counterfactual therefore incorporates the feedback from wages to employment policies throughout the exercise.

Table 7 reports the baseline counterfactual results. Matching the two targets requires raising

the kink parameter by a factor of 1.54 and the quadratic parameter by 2.94. Inaction rises by 6.59 points, against 6.51 in the data, and gross flows fall by 3.56 points, against 3.53. The higher kink raises the cost of each job added or cut, and the higher quadratic parameter makes marginal cost rise more steeply with the size of a change. Establishments adjust less, yet their adjustment expenditures rise from 4.1% to 5.0% of the wage bill. The cost per job reallocated at continuing establishments rises from 0.40 to 0.78 quarterly wages. We then measure annual reallocation from the simulated histories and compute the labor share from the new wage, employment, and output. Because annual reallocation also depends on reversals of quarterly changes within the year, matching the two quarterly targets does not determine its decline.

Panel B of Table 7 reports the untargeted predictions. The labor share falls by 0.95 percentage points and annual reallocation by 5.88 points, accounting for about 24% and 68% of their observed declines. The conditional share of 1–4-job gains rises from 81.3% to 86.2%, compared with 82.4% to 85.5% in the data. The 5–19 share and the 50+ tail both fall. The model thus connects the aggregate declines to more inaction and a greater concentration of small adjustments, although the intermediate 20–49 bin rises.[22]

Average establishment size rises by 1.4%, compared with the BDS increase of 2.2%.[23] Employment-weighted inaction rises by 4.9 percentage points, compared with 2.3 points in the data. Matching the fraction of establishments with unchanged employment does not determine how much employment they account for. Establishment entry changes little, whereas the observed entry rate falls by 1.4 percentage points.

The entry rate and the mass of entrants play different roles in interpreting these outcomes. Free entry pins down the wage: when adjustment costs rise, the wage falls until an entrant again just covers its setup cost. Labor-market clearing then determines the mass of entrants and the scale of the establishment population. The entry rate depends on entrant flows relative to the stock of establishments, which in a stationary economy also reflects the exit and survival of incumbents, so its near-constancy is an outcome of the counterfactual policies and distribution. The model therefore

[22]The shift toward small adjustments also appears when employment changes are weighted by the number of jobs. Appendix Table D.1 shows that the share of gross job gains and losses attributable to changes of 1–4 jobs rises by 7.3 percentage points in the model and 8.0 points in the data.

[23]QCEW average establishment size falls by 10.1%, with establishment counts growing faster than employment. Appendix Table B.8 reports this comparison and the greater count growth in QCEW's smallest size class.

produces a large decline in reallocation with little change in establishment turnover, and the cost increase reaches the labor share mainly through the wage.

We also compare the change in pay relative to output per worker. Appendix Table B.8 shows that real QCEW annual pay per employee grew by 20.6 log points over 2001–2019, compared with 31.0 for BLS output per worker, a gap of -10.4 log points. Using BLS compensation and output for the same nonfarm business sector gives a similar gap of -11.1 log points. With the productivity process held fixed, the model generates a gap of -1.7 log points between the wage change and the change in output per worker. The model reproduces the direction of this gap and accounts for part of its size.

Alternative targeting. Our main empirical regressions relate annual reallocation to labor shares, while the bridge regressions connect rising inaction to declines in both. We therefore also ask how much the labor share would fall if rising adjustment costs accounted for the full annual reallocation decline. Starting from the same 2001 benchmark, we increase the kink and quadratic parameters proportionally to match the observed 8.6-point decline. Appendix Table D.2 compares this exercise with the baseline and the data. The labor share falls by 1.47 percentage points, or 37% of its observed decline, compared with 0.95 points, or 24%, in the baseline. The conditional small-adjustment share still increases. Inaction, however, rises by 14.1 points, against 6.5 in the data, and quarterly gross flows fall by 4.9 points, against 3.5. We retain the quarterly targets as our baseline because they constrain both the frequency and the total size of employment adjustments. The baseline and annual-target exercises give contributions of 24–37%, with the labor share untargeted in both.[24]

6.1. The Mechanism in Detail

We now turn from how much rising adjustment costs lower the labor share to how they lower it. In the baseline counterfactual, the wage and output per worker both fall, but they move the labor share in opposite directions. Both declines start at the firm level. A higher adjustment cost reduces the net gain from changing employment. Some changes cease to be worthwhile, widening the region of inaction. For firms that continue to adjust, the quadratic component raises the marginal cost of

[24] Appendix Table D.3 reports recalibrations and counterfactuals under asymmetric hiring and firing costs, alternative exit assumptions, and numerical settings. The baseline targets the observed decline in jobs reallocated relative to employment, measured in percentage points. Panel F assesses sensitivity to expressing the gross-flow decline relative to its initial rate, holding the inaction target unchanged (Appendix D.3).

large changes and reduces their preferred size. These responses lower reallocation while altering the distribution of employment across productivity states.

The conditional adjustment-size distribution provides a check on these responses. A wider region of inaction removes some small changes, which by itself lowers their share among adjusting firms. A smaller preferred adjustment size shifts remaining changes toward smaller bins. In the baseline counterfactual, the share of gains of 1–4 jobs rises by 4.86 percentage points, the 5–19 share falls by 4.91 points, and the 50+ share falls by 0.43 points. Whether small adjustments become more prevalent therefore depends on the shape of the cost schedule.

Free entry explains the fall in the wage. First hold the wage at its benchmark value. Higher costs reduce the value of a given sequence of future employment decisions. Firms respond by changing their policies, but costlier adjustment lowers the expected value of entry even though initial staffing itself is exempt. At the old wage, this value is below the unchanged entry cost. A lower wage raises operating profits and restores free entry.[25] The adjustment cost is paid when employment changes, but the lower wage applies to every job in every period. At a given output per worker, the lower wage reduces the labor share. Panel C of Table 7 shows that the wage decline contributes -2.67 percentage points to the labor-share change.[26]

The allocation response operates at the same time. After a favorable productivity shock, a firm may expand less than it otherwise would. After an unfavorable shock, another may retain more workers. The new stationary distribution reflects these policy changes, endogenous exit, and the response to the lower wage. Output per worker falls even though the productivity process is held fixed. At a given wage, lower output per worker raises the wage bill relative to output. Panel C reports this offsetting allocation component as $+1.72$ percentage points. This offsets about two-thirds of the wage contribution, leaving a net labor-share decline of 0.95 percentage points.

[25] Appendix Table D.4 measures the fall in entry value when costs rise at the initial wage, allowing establishments to change their policies. It then uses the sensitivity of entry value to wages to approximate the wage decline needed to restore free entry.

[26] Using the benchmark and counterfactual values, the Shapley decomposition averages each component's contribution across two calculation orders: changing the wage first and then employment per unit of output, and changing them in the reverse order. This assigns half of their interaction to each component, so the wage and allocation contributions sum exactly to the labor-share change. Appendix D.4 gives the formulas used in Panel C of Table 7.

6.2. *Discussion*

We next ask what kind of cost increase fits the observed adjustment patterns, and whether the labor share can distinguish rising adjustment costs from smaller productivity shocks.

The roles of the cost components. The relative changes in inaction and gross flows discipline the kink and quadratic parameters jointly. The quadratic cost reduces gross flows more strongly relative to its effect on inaction than the kink. Their joint increase matches both changes. Appendix Figures D.1 and D.2 show the component responses and their implications for the joint changes.

These comparisons hold the benchmark economy fixed and vary the source of the cost increase. Raising the kink makes every job added or cut more costly, while raising curvature makes large changes disproportionately costly. The quadratic term has a strong effect on the size of the changes that remain worthwhile. The kink helps reconcile this response with increased inaction. Another combination is also consistent with the two targets. Raising fixed and quadratic costs while holding the kink unchanged explains 25% of the labor-share decline and 69% of the annual reallocation decline (Appendix Table D.5). The evidence thus supports rising costs with a quadratic component, while the two quarterly changes alone do not select a unique combination of cost increases.

A rise in the fixed cost alone produces a different pattern. The fixed-cost search in Appendix Table D.2 falls short of the annual reallocation target and lowers the conditional small-adjustment share. A higher fixed cost encourages firms to wait and combine employment changes, whereas a higher quadratic cost discourages large adjustments.

In Fujita and Ramey (2007), rising marginal creation costs encourage gradual creation of persistent job positions. Our quadratic term applies the same idea of rising marginal costs to employment changes within establishments. The fixed-cost comparison isolates employment adjustment in our framework. It does not evaluate their full vacancy-creation model, which includes matching and positions that remain available when vacant.

Contribution relative to Decker et al. (2020). A decline in reallocation can come from smaller shocks or from weaker responses to the shocks firms face. Decker et al. (2020) separate these explanations and find that responsiveness to productivity shocks has declined. Rising adjustment costs are one source of weaker responses. Smaller shocks would also raise inaction, so we compare the explanations in our model and examine their implications for the labor share. We separately

adjust the cost parameters and the persistent and transitory shock dispersions to match the rise in quarterly inaction and decline in gross flows. Both mechanisms reduce annual reallocation. Rising costs lower the labor share by 0.95 points, while smaller shocks raise it by 1.25 points. The labor-share response provides an additional way to distinguish the mechanisms.

Appendix Table D.6 reports both counterfactuals. Matching the two targets with shocks requires lowering transitory dispersion by 58 percent and persistent innovation dispersion by 19 percent. The productivity process retains its unconditional mean in levels, and the entrant productivity support is held fixed. Smaller shocks reduce the need to adjust inherited employment. Higher costs instead make it more expensive to close a given gap between inherited and desired employment. With smaller shocks, output per worker falls proportionately more than the wage, raising $w/(Y/L)$. Higher costs reduce firm value at a given wage, and the free-entry wage falls proportionately more than output per worker. Thus, in this comparison, higher adjustment costs account for declining reallocation together with a falling labor share, whereas smaller shocks predict a rising labor share.

The model's contribution is this quantitative link from firm adjustment behavior to factor shares. It offers a common explanation for part of the declines in reallocation and the labor share, with the conditional adjustment-size distribution providing a further test.

7. Conclusion

Rising establishment inaction and smaller employment adjustments accompany the secular declines in U.S. job reallocation and the labor share. Industry-level data across U.S. states show a positive association between reallocation and payroll shares, with larger increases in inaction accompanying larger declines in both. A quarterly model with fixed, kinked, and quadratic adjustment costs links these patterns to the labor share through the equilibrium wage. Increasing the kink and quadratic costs to match quarterly inaction and gross flows explains 24% of the labor-share decline and 68% of the annual reallocation decline. The share of small adjustments rises without being targeted. Matching the full annual reallocation decline instead gives a labor-share contribution of 37%. A fixed-plus-quadratic cost increase produces similar aggregate effects under the two quarterly targets, supporting the role of curvature while allowing different combinations of rising costs.

The results connect declining establishment responsiveness to factor shares and complement

other explanations of the labor-share decline. Future work could identify the sources of rising adjustment costs and examine their interaction with capital substitution and the choice between hours and headcount adjustment.

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Online Appendix for
"The Rise of Inaction: Job Reallocation and the Declining Labor Share"

Appendix A. Data Appendix

This appendix provides detailed descriptions of the data sources, variable construction, and sample restrictions used in the empirical analysis.

Appendix A.1. BEA Regional Economic Accounts (Payroll Share)

The numerator and denominator of our payroll share measure are drawn from the Bureau of Economic Analysis (BEA) Regional Economic Accounts, specifically:

- **SAGDP2N** (GDP in current dollars by state and industry): Provides gross domestic product at the state-by-industry level. The data are reported annually in millions of dollars.
- **SAGDP4N** (Compensation of employees by state and industry): Provides total compensation of employees (wages, salaries, and supplements) at the state-by-industry level. The data are reported annually in thousands of dollars.

The data cover all 50 states plus the District of Columbia, for NAICS-based industry sectors, from 1997 to 2024.[27]

Industry mapping: The BEA reports data using "LineCode" identifiers that correspond to NAICS-based sectors. We map these to our 11 private-sector industry groups as follows:

- LineCode 11: Construction (NAICS 23)
- LineCode 12: Manufacturing (NAICS 31–33)
- LineCode 34: Wholesale Trade (NAICS 42)
- LineCode 35: Retail Trade (NAICS 44–45)
- LineCode 36: Transportation and Warehousing (NAICS 48–49)
- LineCode 45: Information (NAICS 51)
- LineCode 50: Finance, Insurance, Real Estate (NAICS 52–53)
- LineCode 59: Professional and Business Services (NAICS 54–56)

[27] We download the SAGDP data from the BEA website (https://apps.bea.gov/regional/downloadzip.htm).

- LineCode 68: Education and Health Services (NAICS 61–62)
- LineCode 75: Leisure and Hospitality (NAICS 71–72)
- LineCode 82: Other Services (NAICS 81)

We exclude Agriculture (NAICS 11), Mining (NAICS 21), Utilities (NAICS 22), and Government from the state-industry analysis. The payroll share is computed as the ratio of compensation to GDP for each state-industry-year cell.

For the wages-only robustness check (Appendix B.3), we additionally use **SAINC7N** (Wages and salaries by NAICS industry), which reports the wages-and-salaries component of compensation—excluding employer-provided supplements such as health insurance contributions, pension contributions, and payroll taxes—at the state-by-industry level in thousands of dollars. Because SAINC7N reports finer industry detail than the SAGDP supersectors (e.g., Finance (NAICS 52) and Real Estate (NAICS 53) are separate), we sum the component NAICS industries to match our 11 supersector groups before dividing by SAGDP2N GDP.

Appendix A.2. Census Business Dynamics Statistics (Job Reallocation)

The Business Dynamics Statistics (BDS) is an annual dataset produced by the Census Bureau from the Longitudinal Business Database. The BDS tracks establishments and the firms that own them, and provides tabulations of job creation, job destruction, and establishment births and deaths by state, sector, firm age, and firm size. Coverage spans 1978–2023 and includes virtually all private-sector employer firms in the United States.

We use the state-by-sector BDS tables as our primary source for job reallocation. For each state-sector-year cell, the BDS reports the job creation rate (employment gains from expanding and opening establishments as a share of average employment), the job destruction rate (employment losses from contracting and closing establishments), and the job reallocation rate as their sum. We aggregate 2-digit NAICS sectors to 11 private-sector industry groups matching the BEA classification by summing job flows and their average-endpoint employment denominators within each group.

Appendix A.3. BLS Business Employment Dynamics (Size-of-Change Data)

The establishment-level size-of-change analysis in Section 4.1 requires data on whether individual establishments changed employment and, if so, by how much. These data are available only from the BLS Business Employment Dynamics (BED) program, which tracks *establishments* using the Quarterly Census of Employment and Wages (QCEW). We use three BED data products:

- **Establishment no-change data**: The number of private-sector establishments reporting zero net employment change from the prior quarter, available quarterly from 1992Q3 to 2025Q2.

The same tabulation reports total (third-month) employment at no-change establishments, which is the numerator of the employment-weighted inaction variant in Section 4.1.

- **Size-of-change tabulations**: Establishment counts and employment flows classified by the magnitude of the employment change (1 job, 2 jobs, ..., 100+ jobs, as well as broad categories 1–4, 5–19, 20+). Available quarterly from 1992Q3 to 2025Q2, not seasonally adjusted. The granular classes (including 50–74, 75–99, and 100+ jobs) let us construct the conditional 50+ share used as a calibration target in the model, in addition to the 1–4 and 20+ shares. Quarterly gross job gains and losses (as a share of employment), which we use to discipline the transitory-shock volatility, are taken from the BLS's seasonally adjusted series. The no-change and conditional-share moments are not seasonally adjusted, and the three-factor decomposition of Appendix B.5 uses not-seasonally-adjusted levels throughout so that its accounting identity holds exactly.

- **Establishment gaining, losing, opening, and closing rates**: The quarterly rates by state and supersector share an average-endpoint establishment denominator. We convert them to the unchanged share of the endpoint union using $q_U = 100(b+t)/(100+t)$, where $b = 100 - \text{gain rate} - \text{loss rate}$ and $t = (\text{opening rate} + \text{closing rate})/2$. The state-industry proxy averages this share across quarters. The annual March-to-March no-change share uses the same conversion with the published gaining, losing, opening, and closing rates.

The inaction rate is computed as:

$$\text{Inaction Rate}_t = \frac{\text{Establishments with zero change}_t}{\text{Establishments with zero change}_t + \text{Establishments with any change}_t}$$

where the denominator sums the establishments with zero change and those with gains (across all size categories) and those with losses (across all size categories).[28]

The BED defines zero change as exact equality of reported employment in the third month of consecutive quarters, using the pay period that includes the 12th. This is a measure of zero *net* change: offsetting hires and separations or adjustments reversed between observation dates can leave employment unchanged. Conversely, a temporary change at an observation date can register as adjustment. Reporting errors can move establishments into or out of the zero-change category. BLS links the underlying QCEW records across quarters and revises the published series for corrections to employment reports and establishment links (see the BED technical notes). The quarterly model uses the same net-change definition.

[28] These data are sourced from https://www.bls.gov/bdm/bdsoc.htm.

Two alternative measures show that the secular rise survives changes in frequency and weighting. The annual March-to-March no-change rate, constructed from the BED's annual gaining, losing, opening, and closing rates, rises from 29.8% in 2001 to 36.4% in 2019. Employment-weighted quarterly no-change rises from 14.3% to 16.6%. For this rate, we divide employment at no-change establishments by BED average-endpoint employment reconstructed from NSA flow levels and rates. The reconstruction accounts for the rounding of published rates. Annual net change can include within-year reversals, so the quarterly and annual rates measure distinct adjustment outcomes.

Appendix A.4. BLS Quarterly Census of Employment and Wages (QCEW)

The QCEW is the quarterly administrative census of establishments covered by state unemployment-insurance programs—the source records from which the BED is constructed—reporting employment by month, establishment counts, and total wages. We construct national private-sector average establishment size and average annual pay in 2001 and 2019 from the 2001 QCEW annual singlefile and the 2019 annual national data (BLS CEW open-data API). Average size is annual average employment divided by annual average establishment counts. Annual pay is total annual wages divided by annual average employment. Counts of establishments by size class come from the 2001 and 2019 QCEW size tabulations, which BLS publishes only for the first quarter, with size based on March employment. These comparisons are reported in Appendix B.8.

For the wage comparison, we supplement QCEW with the BLS nonfarm business series for employment, compensation, real output, and nominal output. All BLS series come from one frozen download retrieved on September 14, 2026. We use published annual observations (period Q05) and form ratios from the annual aggregates. The output deflator is nominal output divided by real output. We deflate QCEW annual pay and BLS compensation by this index, and compare their growth per worker with real output per worker. The replication package records the series identifiers and source vintage. BLS compensation includes employer benefits and imputed compensation of the self-employed. Its employment and output cover the same nonfarm business sector, which excludes farms, general government, private households, and nonprofit institutions serving households. QCEW private employment includes some of these activities and excludes the self-employed. The matched BLS comparison therefore supplements the QCEW earnings evidence. Its sector and self-employment coverage also differ from the paper's BEA employee-compensation benchmark.[29]

We also build a continuous series of national private third-month employment from 1992 to 2025, using quarterly singlefile extracts through 2000 and the BLS timeseries API thereafter. It supplies the employment denominator for the three-factor decomposition in Appendix B.5 and an alternative employment-weighted no-change rate. The QCEW-based rate rises from 14.3% to

[29] See the BLS productivity concepts and calculation methods. We use the output deflator to express compensation in units of the model's final good.

16.5%, close to the BED-based rate used in the calibration. QCEW private-sector coverage is broader than the BED universe, so we retain the BED-based denominator for the model comparison. All extracts are frozen in the replication package.

Appendix A.5. Census SUSB (Market Concentration)

The Statistics of U.S. Businesses (SUSB) is an annual dataset produced by the Census Bureau from administrative records. It reports the number of firms and total employment by enterprise employment-size class for each state-industry cell. Size classes include 1–4, 5–9, 10–19, 20–49, 50–99, 100–249, 250–499, 500–999, 1000–2499, 2500–4999, 5000–9999, and 10000+ employees.

We compute the employment-based Herfindahl-Hirschman Index (HHI) using these size classes as a proxy for employment concentration. Because the SUSB reports firm sizes in groups, we approximate the firm-level HHI by dividing each bin's squared employment share by the number of firms in that bin (Equation 3), assuming equal-sized firms within each bin. While this remains an imperfect measure of true product-market concentration, it correctly accounts for the number of firms per size class and provides a consistent control for market structure at the state-industry-year level. The SUSB data are available from 1998 to 2022. We also use SUSB data to compute the manufacturing and goods-producing employment shares at the state-year level.

Appendix A.6. BDS Establishment Entry Rate

We use the establishment entry rate reported directly in the BDS. The BDS defines establishment entries as new establishments appearing in the Longitudinal Business Database in a given year, and reports the entry rate as the number of entering establishments divided by the average number of employment-active establishments at the current and preceding annual endpoints, using a consistent scope across the two years. This yields entry rates of 10.99% for 2001 and 9.61% for 2019. The same BDS aggregates give average establishment size as total private-sector employment divided by the total establishment count: 17.97 employees in 2001 (reported as 18.0 in Tables 6 and 7) and 18.37 in 2019. The 2001 value is the establishment-size calibration target.

Appendix A.7. Union Membership Data

State-level private-sector union membership rates are obtained from the Union Membership and Coverage Database maintained by Hirsch and Macpherson (2003), available at `https://unionstats.com`. These estimates are based on the Current Population Survey (CPS) and provide consistent annual coverage for all 50 states plus the District of Columbia from 1983 to 2025. We use the private-sector union membership rate (percentage of private-sector workers who are union members), which varies at the state-by-year level. The data are merged into our panel by state FIPS code and year, providing a control for the bargaining-power channel of the labor share decline.

Appendix Table A.1: Summary Statistics

Panel A: Full Sample (2001–2019)

Variable	Mean	Std.Dev.	Min	Median	Max	Obs.
Payroll share (%)	56.63	16.85	12.21	59.28	90.93	10631
Job reallocation rate (%)	27.46	7.67	10.49	26.52	74.65	10631
Job creation rate (%)	13.97	4.20	4.28	13.44	41.07	10631
Job destruction rate (%)	13.49	4.57	4.41	12.85	61.32	10631
HHI (employment)	0.0019	0.0027	0.0001	0.0011	0.0625	10631
Union membership (%)	6.59	3.46	0.81	5.82	16.91	10631
Mfg. employment share	0.11	0.05	0.00	0.11	0.26	10631

Panel B: Means by Year

Variable	2001	2005	2010	2015	2019
Payroll share (%)	60.22	56.72	55.41	55.52	55.36
Job reallocation rate (%)	31.58	28.64	26.80	24.16	22.73
Job creation rate (%)	15.98	14.84	11.85	13.14	12.14
Job destruction rate (%)	15.60	13.80	14.94	11.02	10.58
HHI (employment)	0.0012	0.0020	0.0020	0.0019	0.0020
Union membership (%)	8.22	7.01	6.34	6.07	5.87
Mfg. employment share	0.15	0.12	0.10	0.10	0.10

Panel C: Long-Differences (2001–2019)

Variable	Mean Δ	Std.Dev.	Obs.
Payroll share (p.p.)	-4.81	7.31	551
Job reallocation rate (p.p.)	-8.92	6.20	551
Job creation rate (p.p.)	-3.88	4.46	551
Job destruction rate (p.p.)	-5.04	3.67	551
HHI (employment)	0.0007	0.0012	551
Union membership (p.p.)	-2.34	1.87	551
Mfg. employment share	-0.05	0.02	551

Notes: The table reports summary statistics at the state-industry level. The payroll share is the ratio of compensation of employees to gross value added (%). Job reallocation, creation, and destruction rates are from the Census BDS (%). The HHI is a grouped-data employment concentration proxy constructed from Census SUSB firm counts and employment by size class. The union membership rate covers the private sector and is drawn from unionstats.com (Hirsch & Macpherson). The manufacturing employment share is computed from SUSB data.

Appendix A.8. Summary Statistics

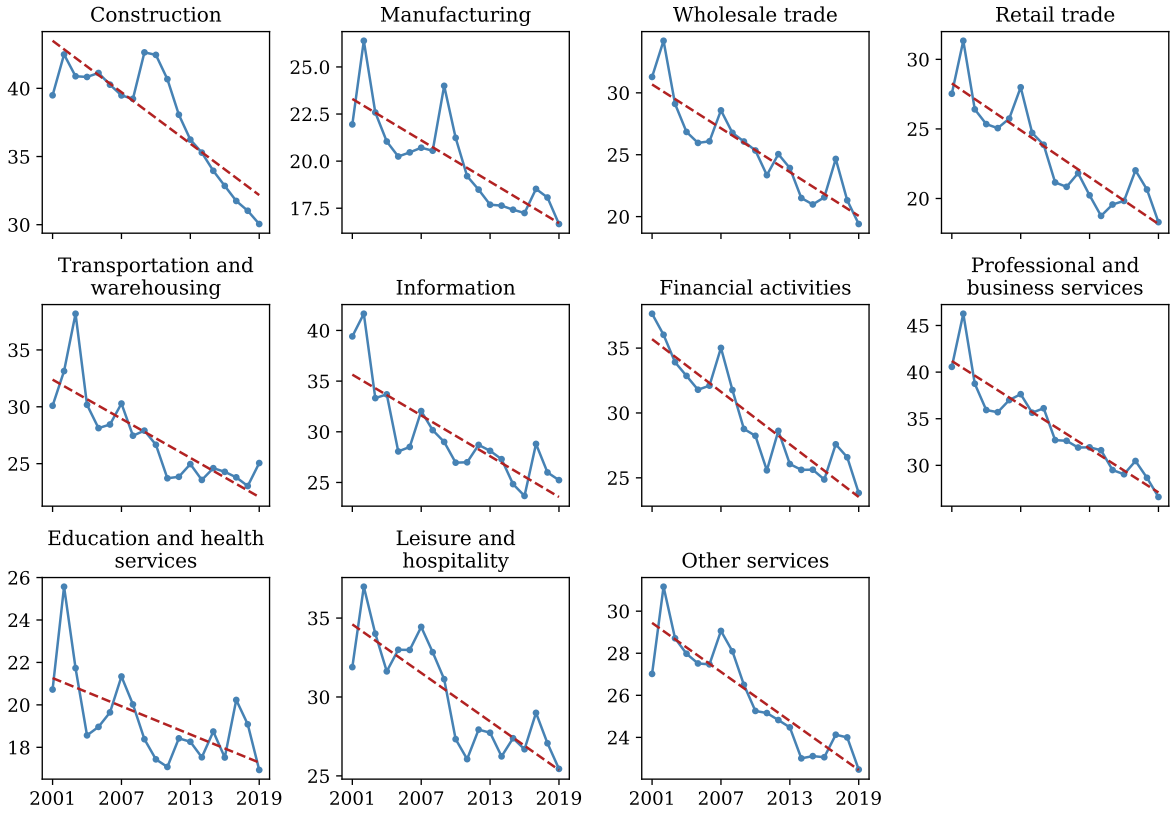
Appendix Table A.1 reports the summary statistics for the state-industry panel used throughout Section 3: the pooled distribution of the payroll share, job reallocation rate, and controls in Panel A. Panel B gives their 2001 and 2019 means, and Panel C reports the distribution of long-differences. The discussion in the main text draws on these moments.

Appendix B. Additional Empirical Results

This appendix reports additional empirical results, in the following order. We begin with industry-level variation in the JRR and payroll share, the temporal stability of the reallocation–payroll-share relationship and the inaction–reallocation bridge, and a wages-only variant of the payroll share. Next come the establishment-level BED results, the exact three-factor decomposition of the gross-flow decline, and the bridge between BED inaction and the BDS panel. We close with a discussion of possible sources of rising adjustment costs and with QCEW comparisons of establishment size and pay per worker.

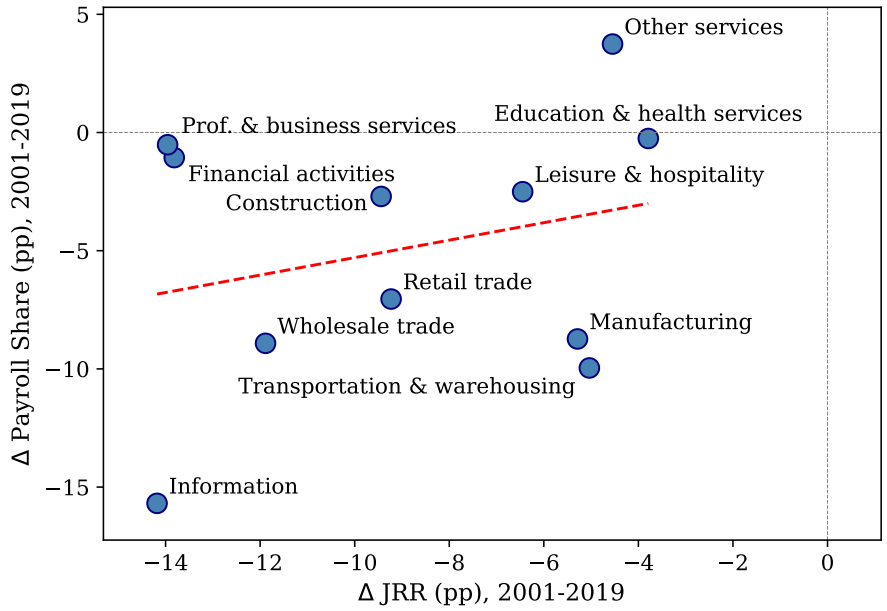
Appendix B.1. Industry-Level Changes

Appendix Figure B.1 plots the JRR separately for each of our 11 industry groups, confirming the universality of the decline noted in Section 2. Every industry declined between 2001 and 2019, with the largest declines in information (−14.2 pp) and professional and business services (−14.0 pp) and the smallest in education and health services (−3.8 pp). Appendix Figure B.2 then plots



Appendix Figure B.1: Job Reallocation Rate by Industry, 2001–2019

Notes: Each panel plots the JRR for one of the 11 industry groups, with a fitted linear trend. The decline in job reallocation is universal across industries. Job reallocation from Census BDS.



Appendix Figure B.2: Industry-Level Changes in JRR and Payroll Share, 2001–2019

Notes: Each point represents one of the 11 industry groups. The x-axis is the change in the average JRR (pp) and the y-axis is the change in the average payroll share (pp), both from 2001 to 2019. The dashed line is a linear fit. Job reallocation from Census BDS.

the industry-level change in the JRR against the change in the payroll share, showing a positive cross-industry correlation that mirrors the within-industry panel relationship.

Appendix B.2. Temporal Stability and the Inaction–Reallocation Bridge

Appendix Table B.1 reports the direct long-difference regressions. Panel A uses 2001–2019 changes, with the full-control specification in column 6 corresponding to Section 3.1. Panel B uses 2001–2007 changes. The dependent variable, JRR, and concentration enter as log differences, while union membership and manufacturing exposure enter as level differences. These estimates describe persistent differences across cells, while the annual fixed-effects regressions can also reflect local cyclical variation. Neither design identifies a causal effect of adjustment costs.

Appendix Table B.2 reports the sub-period estimates discussed in Section 3.2. Its post-recession specification excludes 2008–2009, which are included in the later period of Table 2. The post-recession point estimate (2010–2019) is smaller than the pre-recession estimate (0.017 vs. 0.032). Excluding the immediate recovery years, 2010–2011, brings the 2012–2019 estimate to 0.024 ($p < 0.01$), closer to its pre-recession value. The full-sample coefficient including the COVID period (2001–2022) remains positive and precisely estimated at 0.042. The reported significance levels concern each coefficient relative to zero and do not test equality across periods. We therefore describe the variation without claiming a statistically established change in the relationship.

Appendix Table B.1: Labor Shares and Job Reallocation in Long Differences

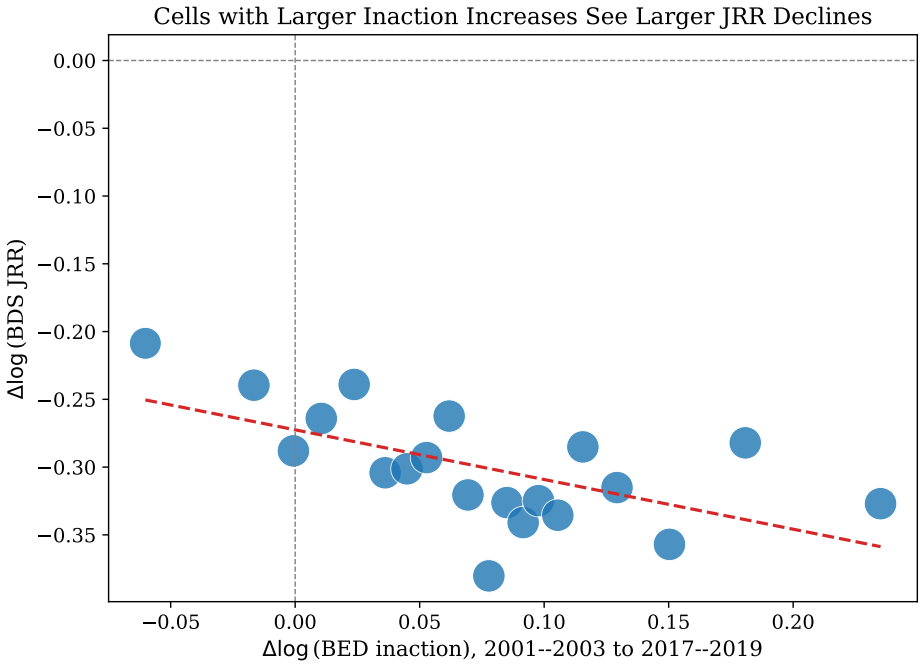
	Changes between period endpoints					
Dep.Var.: Δ log Payroll share	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: 2001–2019						
Δ log Job reallocation	0.160***	0.157***		0.165***	0.158***	0.164***
	(0.030)	(0.032)		(0.033)	(0.032)	(0.033)
Δ log HHI (concentration)		-0.010	-0.026	-0.005	-0.012	-0.006
		(0.022)	(0.021)	(0.023)	(0.022)	(0.023)
Δ Union membership				0.008***		0.007***
				(0.003)		(0.003)
Δ Mfg. share					-0.376	-0.258
					(0.303)	(0.290)
R^2	0.04	0.04	0.00	0.05	0.04	0.05
Observations	551	551	551	551	551	551
Panel B: 2001–2007						
Δ log Job reallocation	0.143***	0.127***		0.126***	0.129***	0.129***
	(0.026)	(0.025)		(0.025)	(0.026)	(0.026)
Δ log HHI (concentration)		-0.080***	-0.100***	-0.081***	-0.079***	-0.080***
		(0.025)	(0.025)	(0.025)	(0.025)	(0.025)
Δ Union membership				-0.002		-0.001
				(0.003)		(0.003)
Δ Mfg. share					-0.588***	-0.585***
					(0.211)	(0.209)
R^2	0.05	0.07	0.03	0.07	0.07	0.07
Observations	551	551	551	551	551	551

Notes: Job reallocation measures establishment job gains and losses from the Census BDS. Changes in the payroll share, job reallocation, and HHI are log differences, while union membership and manufacturing share enter as level differences. All specifications include an intercept. Column (6) includes all three controls. Standard errors are clustered at the state level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix Figure B.3 visualizes the bridge relationship discussed in Section 4.2: state-industry cells with larger increases in establishment-level inaction experienced larger declines in BDS job reallocation, formalized in Table 4.

Appendix B.3. Wages-Only Share Robustness

Our baseline dependent variable is the compensation share (Compensation of Employees / GDP, BEA SAGDP4/SAGDP2), which includes employer-provided supplements such as health insurance contributions. As discussed in Section 4.4, rising healthcare costs mechanically increase the compensation numerator, raising a potential concern that the JRR–labor-share relationship partly



Appendix Figure B.3: State-Industry Changes in BED Inaction and BDS Reallocation

Notes: Each point is a bin of state-industry cells. The x-axis is the change in log BED inaction between 2001–2003 and 2017–2019. The y-axis is the corresponding change in log BDS gross job reallocation. The dashed line is the linear fit across all state-industry cells.
Source: Bureau of Labor Statistics, Business Employment Dynamics. Census Bureau, Business Dynamics Statistics. Bureau of Economic Analysis.

reflects a spurious correlation through the supplements component. To address this directly, we reconstruct the dependent variable using wages and salaries only (BEA SAINC7N), excluding all employer supplements.

Appendix Table B.3 reports the results. In the annual panel specification, the wages-only JRR

Appendix Table B.2: JRR–Labor Share Relationship by Sub-Period

Period	Years	β(JRR)	SE	N
Full pre-COVID	2001–2019	0.045***	(0.008)	10631
Pre-GR	2001–2007	0.032***	(0.011)	3904
Post-GR	2010–2019	0.017*	(0.010)	5605
Post-GR (excl. recovery)	2012–2019	0.024***	(0.009)	4483
Full incl. COVID	2001–2022	0.042***	(0.008)	12308

Notes: Each row reports the coefficient from a panel fixed-effects regression of the log payroll share on the log JRR, controlling for log employment HHI, with state-industry and year fixed effects. Standard errors are clustered at the state-industry level and reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Appendix Table B.3: Robustness: Wages-Only Share vs. Compensation Share

	Cross-section (2001–2019)		Panel (2001–2019)	
	(1)	(2)	(3)	(4)
Panel A: Compensation share (baseline)				
Job reallocation rate	0.160***	0.157***	0.044***	0.045***
	(0.030)	(0.032)	(0.008)	(0.008)
HHI (concentration)		-0.010		0.012
		(0.022)		(0.013)
R^2	0.04	0.04		
R^2 (within)			0.050	0.043
Observations	551	551	10631	10631
Panel B: Wages-only share (excl. supplements)				
Job reallocation rate	0.160***	0.157***	0.044***	0.044***
	(0.032)	(0.034)	(0.008)	(0.008)
HHI (concentration)		-0.010		0.007
		(0.023)		(0.013)
R^2	0.04	0.04		
R^2 (within)			0.046	0.042
Observations	551	551	10631	10631
HHI control	No	Yes	No	Yes
Entity FE	No	No	Yes	Yes
Year FE	No	No	Yes	Yes

Notes: Panel A uses the compensation share (compensation of employees divided by GDP, BEA SAGDP4/SAGDP2), which includes employer-provided supplements such as health insurance contributions. Panel B uses the wages-only share (wages and salaries divided by GDP, BEA SAINC7N/SAGDP2), which excludes supplements. Job reallocation from Census BDS. Columns (1)–(2) report cross-sectional long-difference regressions (log changes, 2001–2019). Standard errors are clustered at the state level. Columns (3)–(4) report panel regressions with state×industry and year fixed effects. Standard errors are clustered at the entity level. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

coefficient is essentially unchanged from the compensation-based coefficient. The positive annual association is therefore also present for wages alone and does not depend on including employer supplements in compensation.

Appendix B.4. BED Establishment-Level Results

Our main analysis uses the Census Bureau's Business Dynamics Statistics (BDS), which builds annual job flows from the Longitudinal Business Database. As a robustness check, we replicate the core panel results using the BLS BED, which builds quarterly job flows from the QCEW establishment records. The BED regressor is the calendar-year mean of quarterly job

Appendix Table B.4: Panel Regression Results (BED Establishment-Level Data)

	log(Payroll share)					
	(1)	(2)	(3)	(4)	(5)	(6)
Job reallocation rate	0.050**	0.049**		0.049**	0.049**	0.049**
	(0.020)	(0.020)		(0.020)	(0.020)	(0.020)
HHI (concentration)		-0.004	-0.009	-0.003	-0.004	-0.003
		(0.013)	(0.014)	(0.013)	(0.013)	(0.013)
Union membership				0.003*		0.003*
				(0.001)		(0.001)
Mfg. share					-0.211	-0.199
					(0.241)	(0.241)
Observations	9473	9473	9473	9473	9473	9473

Notes: Dependent variable: log payroll share. Sample period 2001–2019. Job reallocation is the calendar-year mean of quarterly BLS BED establishment-level rates. Column (1) includes only JRR. Column (2) adds HHI. Column (3) HHI only. Column (4) adds union rate. Column (5) adds mfg. share. Column (6) all controls. All specifications include state×industry and year FE. Standard errors clustered at state×industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

reallocation rates, whereas the BDS regressor uses March-to-March annual flows. These observation windows measure different employment changes. The BED provides finer establishment-level detail, including the size-of-change data used for the inaction analysis (Figure 3), and measures reallocation between establishments. Appendix Table B.4 reports the BED panel regression results. The JRR–labor-share relationship is qualitatively similar, with the BED yielding a slightly larger baseline coefficient.

Appendix B.5. An Exact Decomposition of the Gross-Flow Decline

This appendix quantifies the frequency, relative-size, and conditional-intensity components of the observed decline in gross job flows. Write the quarterly identity as

$$G_t = \underbrace{\frac{N_t^A}{N_t}}_{\text{share adjusting}} \times \underbrace{\frac{D_t^A / N_t^A}{D_t / N_t}}_{\text{relative size of adjusting}} \times \underbrace{\frac{J_t}{D_t^A}}_{\text{conditional intensity}},$$

where J_t is BED gross job gains plus losses in levels (NSA), D_t is average-endpoint employment from QCEW national private third-month employment, and D_t^{NC} is BED employment at no-change establishments. Put $D_t^A = D_t - D_t^{NC}$ and let N_t and N_t^A denote the union count and adjusting count in the BED size-of-change universe. Because QCEW coverage is broader than BED coverage, D_t^A is a residual exposure proxy, not a directly observed BED employment total for adjusters. The identity

Appendix Table B.5: Three-Factor Decomposition of the Gross-Flow Decline, 2001–2019

Factor	2001	2019	$\Delta \log$	Share of $\Delta \log G$
Gross flows G	0.151	0.117	-0.256	100.0%
Share of establishments adjusting (N^A/N)	0.545	0.480	-0.127	49.7%
Relative size of adjusting establishments	1.574	1.741	$+0.101$	-39.4%
Conditional intensity (J/D^A)	0.176	0.140	-0.229	89.7%

Notes: BED NSA gross gains-plus-losses, no-change employment and size-of-change establishment counts are combined with a QCEW average-endpoint employment denominator. The QCEW denominator is an exposure proxy because the sources differ in coverage. Each annual factor is a ratio of pooled quarterly underlying levels, so the product identity and the sum of log changes hold exactly for these constructed objects. The gross-flow rate is an employment-exposure-weighted mean quarterly rate, distinct from March-to-March annual flows and from the annual mean of seasonally adjusted quarterly rates used in Figure 3. In the equivalent two-factor decomposition $G = (1-e)I$, the employment share of adjusting establishments accounts for 10.3% of the log decline and conditional intensity for 89.7%.

holds exactly for these constructed quantities, although the coverage difference remains.

For each calendar year y, pool the underlying quarterly levels before forming the ratios:

$$G_y = \frac{\sum_{t\in y} J_t}{\sum_{t\in y} D_t} = \frac{\sum_{t\in y} N_t^A}{\sum_{t\in y} N_t} \frac{(\sum_{t\in y} D_t^A)/(\sum_{t\in y} N_t^A)}{(\sum_{t\in y} D_t)/(\sum_{t\in y} N_t)} \frac{\sum_{t\in y} J_t}{\sum_{t\in y} D_t^A}.$$

Appendix Table B.5 reports these factors for 2001 and 2019. Their log changes sum exactly to the log change in G_y. This is an employment-exposure-weighted quarterly flow rate, distinct from both March-to-March annual reallocation and the arithmetic mean of seasonally adjusted quarterly rates in Figure 3. The quarterly no-change series in that figure also retains its calendar-year arithmetic averaging.

The falling share of establishments that adjust accounts for 49.75% of the log gross-flow decline. Rising relative size offsets 39.43%, while lower conditional intensity accounts for 89.68%. Thus, the size-of-adjustment component remains the largest contributor under exact annual pooling. Using the rate-implied BED employment exposure as an alternative gives the same signs and similar shares. These accounting contributions do not separately identify behavioral changes and establishment-composition effects.

Appendix B.6. Bridging BED Inaction and the BDS Panel

Appendix Table B.6 reports bridge regressions by subsample. The continuing-establishment coefficient is larger in magnitude in lower-concentration cells (-0.501 versus -0.267) and similar before and after 2007 (-0.524 and -0.534). The payroll-share coefficient is negative in both goods and services (-0.257 and -0.208). The extensive-margin coefficient remains larger in magnitude than the continuing-establishment coefficient in every reported subsample. The table reports the

Appendix Table B.6: Where Is the Inaction Bridge Strongest?

	log JRR	log intensive JRR	log extensive JRR	log payroll share
Panel A. Full sample				
log(BED inaction)	-0.474***	-0.326***	-0.802***	-0.175***
	(0.062)	(0.055)	(0.107)	(0.050)
N	9473	9473	9473	9473
Panel B. Goods				
log(BED inaction)	-0.609***	-0.398***	-1.222***	-0.257**
	(0.116)	(0.107)	(0.245)	(0.126)
N	1687	1687	1687	1687
Panel C. Services				
log(BED inaction)	-0.382***	-0.218***	-0.644***	-0.208***
	(0.075)	(0.065)	(0.129)	(0.058)
N	7786	7786	7786	7786
Panel D. Pre-GR (2001-2007)				
log(BED inaction)	-0.720***	-0.524***	-1.039***	-0.529***
	(0.135)	(0.124)	(0.227)	(0.100)
N	3486	3486	3486	3486
Panel E. Great Recession and afterward (2008-2019)				
log(BED inaction)	-0.655***	-0.534***	-0.924***	0.132**
	(0.090)	(0.086)	(0.136)	(0.053)
N	5987	5987	5987	5987
Panel F. Low HHI				
log(BED inaction)	-0.651***	-0.501***	-1.041***	-0.170**
	(0.074)	(0.067)	(0.131)	(0.067)
N	4712	4712	4712	4712
Panel G. High HHI				
log(BED inaction)	-0.400***	-0.267***	-0.680***	-0.171**
	(0.088)	(0.077)	(0.155)	(0.069)
N	4689	4689	4689	4689

Notes: Each entry reports the coefficient on log(BED inaction) from a panel fixed-effects regression with state-industry and year fixed effects. The dependent variable changes across columns. Subsamples are defined by industry type, time period, and 2001 median HHI. Standard errors are clustered by state-industry cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

estimates without testing equality across subsamples.

Appendix Table B.7 compares definitions and coverage on a balanced sample of 492 state-industry cells, using the same controls in the annual and long-difference specifications. This common sample is smaller than those used in the main tables because it requires all outcomes,

Appendix Table B.7: Inaction bridge: matched-sample sensitivity checks

Inaction measure	Baseline q_U	Excluding FIRE q_U	Continuers q_C	With turnover q_U
A. Annual panel, 2001–2019				
Payroll share	-0.168***	-0.177***	-0.077*	-0.116***
	(0.041)	(0.043)	(0.040)	(0.042)
Total JRR	-0.444***	-0.440***	-0.201**	-0.304***
	(0.068)	(0.070)	(0.080)	(0.066)
Continuing-establishment JRR	-0.308***	-0.313***	-0.107	-0.190***
	(0.064)	(0.065)	(0.071)	(0.061)
Birth/death JRR	-0.745***	-0.727***	-0.430***	-0.570***
	(0.107)	(0.110)	(0.128)	(0.112)
JRR coefficient: payroll share	0.054***	0.060***	—	—
	(0.007)	(0.008)		
Observations	9,348	8,436	9,348	9,348
State–industry cells	492	444	492	492
B. Long differences, 2001–03 to 2017–19				
Payroll share	-0.303***	-0.335***	-0.208***	-0.290***
	(0.064)	(0.065)	(0.064)	(0.066)
Total JRR	-0.363***	-0.349***	-0.248**	-0.348***
	(0.094)	(0.096)	(0.099)	(0.093)
Continuing-establishment JRR	-0.138	-0.121	-0.060	-0.127
	(0.092)	(0.094)	(0.090)	(0.088)
Birth/death JRR	-0.847***	-0.845***	-0.660***	-0.826***
	(0.169)	(0.169)	(0.177)	(0.174)
JRR coefficient: payroll share	0.215***	0.251***	—	—
	(0.055)	(0.057)		
Observations	492	444	492	492
State–industry cells	492	444	492	492

Notes: Entries report log-inaction coefficients in separate ordinary least-squares regressions with the indicated log outcome. The final coefficient row instead regresses log payroll share on log total JRR. All regressions include log employment HHI, state union membership (percentage points), and the state manufacturing employment share (fraction). Panel A includes state–industry and year fixed effects. Panel B uses log changes of arithmetic endpoint means and corresponding changes in controls, with an intercept. The baseline, continuer, and turnover columns use the same cells with all outcomes, inaction measures, and controls observed in every year. The FIRE column removes financial activities from this sample. Standard errors in parentheses are clustered by state. All columns have 51 clusters, and significance uses a t_{50} reference distribution. q_U is the unchanged share of the union of establishments active at either endpoint, while q_C uses establishments active at both endpoints. Both are reconstructed quarterly before annual averaging. The turnover control is the annual mean of one-half the sum of BED opening and closing establishment rates, measured in percentage points relative to the average endpoint establishment count. Panel B uses its endpoint change. Conditioning on turnover is descriptive. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

inaction definitions, and controls in every year. Long differences compare 2001–2003 averages with 2017–2019 averages. Standard errors are clustered by state.

The first column uses the union-based no-change share q_U. The second excludes financial

activities (FIRE), leaving 444 cells. This checks sensitivity to a sector whose GDP includes owner-occupied housing services without corresponding employer job flows.[30] The positive direct JRR–payroll-share association and negative inaction–payroll-share association remain in both horizons.

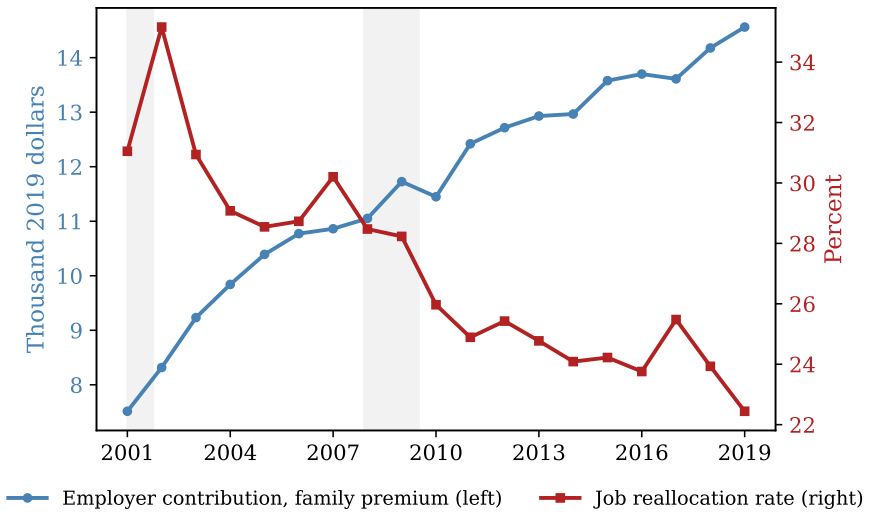
The third column replaces q_U with inaction conditional on continuing establishments. With the definitions of b and t in Section 4.2, this share is $q_C = 100(b+t)/(100-t)$, reconstructed quarterly before annual averaging. The annual payroll-share coefficient becomes smaller and less precise (-0.077, $p = 0.061$), while the long-difference coefficient remains negative and significant (-0.208). The fourth column retains q_U and controls for the average BED opening and closing establishment rates. Its payroll-share coefficients remain negative (-0.116 annually and -0.290 in long differences), as do the birth/death-JRR coefficients. The continuing-establishment-JRR long differences remain imprecisely estimated across these comparisons. Turnover may itself respond to adjustment frictions, so these are descriptive conditional checks rather than estimates of a turnover-independent causal mechanism.

Appendix B.7. Possible Sources of Rising Adjustment Costs

Regulatory and organizational costs could contribute to the effective adjustment friction. Trebbi et al. (2026) estimate that regulatory compliance absorbs 1.3–3.3% of the average firm's wage bill and document rising compliance costs over 2002–2014. Workforce changes can require additional administrative and organizational work. Advance-notice requirements for large layoffs provide another example of costs that depend on the size of an employment change.

Direct measurements put hiring costs at several weeks of wages per hire. For skilled workers in Swiss firms, average hiring costs are 10 to 17 weeks of wages depending on firm size, with no evidence of a fixed component (Blatter et al., 2012). In German firms they exceed eight weeks of wages, and the cost per hire rises with the number of hires (Muehlemann and Pfeifer, 2016). For a Chinese clothing retailer, ten percent higher turnover costs about as much as a 0.6 percent wage increase (Kuhn and Yu, 2021). This estimate provides evidence on the level of turnover costs, rather than their trend in the United States. Advance-notice rules make part of the cost of large reductions a function of their size. The federal WARN Act of 1988 requires employers with 100 or more employees to give 60 days' notice of a plant closing that affects 50 or more workers at a site and of a mass layoff that affects at least 50 workers and a third of the site's workforce, or 500 or more workers regardless of share. Addison and Blackburn (1994) find that the Act raised the provision of notice only modestly, but during our sample several states adopted their own laws with lower thresholds or longer notice: California (2003, employers with 75 or more employees), Illinois

[30] See the BEA explanation of real-estate GDP. Excluding the entire sector does not isolate the contribution of owner-occupied housing.



Appendix Figure B.4: Employer Health-Premium Contributions and Job Reallocation, 2001–2019

Notes: The left axis plots the average annual employer contribution to the family health insurance premium from the KFF Employer Health Benefits Survey, deflated with the CPI-U to 2019 dollars. The right axis plots the national BDS job reallocation rate, as in Figure 1. Shaded areas indicate NBER recession dates. The figure is descriptive: it places two national trends side by side and does not identify a link between them.

(2005, 75 or more), New Jersey (2007, 100 or more), and New York (2009, 50 or more employees and 90 days' notice). Noncompete agreements and occupational licensing provide further examples of mobility constraints: about 18 percent of labor force participants are bound by a noncompete agreement and 38 percent have signed one at some point (Starr et al., 2021), and the share of workers holding an occupational license rose from under 5 percent in the 1950s to 29 percent by 2008 (Kleiner and Krueger, 2013). These statistics document prevalence and a longer-run licensing trend, rather than a measured increase over 2001–2019.

Employer-provided benefits add per-worker tasks at hiring and separation. Adding a worker requires enrolling the worker in health and other benefit plans, and a separation requires terminating coverage and issuing the continuation-coverage notices that COBRA requires of employers with 20 or more employees. The benefit package these tasks administer has grown over the sample. In the Kaiser Family Foundation (KFF) Employer Health Benefits Survey, conducted annually since 1999, the average annual employer contribution to family health insurance premiums rises from approximately $7,500 to $14,600 between 2001 and 2019, in 2019 dollars deflated with the CPI-U. Appendix Figure B.4 plots this series against the job reallocation rate.

Appendix B.8. QCEW Establishment Size and Pay per Worker

The BED size-of-change tabulations do not directly report average establishment size. We construct the requested QCEW comparison and also reconstruct a BED-based size proxy from published employment levels, rates, and active-establishment counts. We compare it with annual

Appendix Table B.8: QCEW moments, 2001–2019: size and wage comparisons

	2001	2019	Change
Panel A: Average establishment size (employees)			
QCEW	14.1	12.7	−10.1%
BDS	18.0	18.4	+2.2%
Model (baseline counterfactual)	18.9	19.1	+1.4%
Employment and establishment-count comparison			
Establishment count growth	QCEW +28.6% vs. BDS +13.3%		
Employment growth	QCEW +15.6% vs. BDS +15.9%		
Employment growth / establishment-count growth (levels)	QCEW 7.7 vs. BDS 21.4 employees		
Smallest size class (0–4) count growth	QCEW +36.2% vs. BDS (1–4) +14.8%		
Largest size class (1,000+) count growth	QCEW +5.1% vs. BDS +22.2%		
QCEW size at BDS establishment growth	14.1 → 14.4 (+2.0%)		
Panel B: Wages and output per worker (log-point changes)			
QCEW real annual pay per employee			+20.6
BLS real output per worker			+31.0
QCEW pay minus BLS output per worker			−10.4
BLS real compensation per worker			+19.9
BLS compensation minus output per worker			−11.1
Panel C: Model wage response (log-point changes)			
Equilibrium wage			−4.7
Output per worker			−3.0
Wage minus output per worker			−1.7

Notes: Panel A uses QCEW national private totals and Census BDS. The employment-growth/establishment-count-growth row is a ratio of aggregate changes, not entrant size. Panel B uses annual observations from a common BLS productivity-data vintage. BLS compensation, employment, and output all refer to nonfarm business. Compensation includes benefits and imputed compensation of the self-employed. Real pay and compensation use the output deflator. Dividing compensation and output by the same employment measure gives a growth gap equal to the log change in the BLS labor-share index. The QCEW pay comparison uses the same per-worker productivity concept, with differences in compensation and sector coverage retained.

Census BDS, whose 2001 size is the calibration target alongside annual reallocation and entry (Section 5.3). Average size is employment divided by establishment counts in both sources. Panel A of Appendix Table B.8 compares their 2001–2019 changes with the model. Average establishment size falls by 10.1% in QCEW and rises by 2.2% in BDS, compared with 1.4% in the model. The divergence is concentrated in establishment counts: QCEW counts grow by 28.6%, against 13.3% in BDS, while employment grows by 15.6% and 15.9%, respectively. The size-class breakdown locates much of the additional QCEW count growth in small establishments: counts in the QCEW 0–4-employee category grow by 36.2%, compared with 14.8% in the BDS 1–4 category. Faster count growth alongside similar employment growth lowers QCEW average size. We retain BDS as the calibration benchmark and report both data sources.

Differences in coverage and worksite reporting can contribute to the QCEW–BDS count

divergence (Cao et al., 2026). They do not establish an explanation for the BED inaction trend: BED excludes private households and establishments with zero employment at both quarterly endpoints.[31] A BED-based size proxy also falls, from 17.00 employees in 2001 to 15.26 in 2019. We reconstruct average-endpoint employment from the published NSA job-flow levels and rates and divide it by the average active-establishment count. The estimate accounts for rounding in the published inputs. The resulting series is a proxy for average establishment size.

The available aggregate and size-class comparisons do not identify how much of rising quarterly inaction reflects changes within size groups versus changes in establishment composition. The tabulations used here do not provide quarterly no-change rates jointly classified by establishment size and the state-industry cells in our panel. We therefore report the size divergence without assigning it entirely to reporting coverage or interpreting the full inaction increase as a measured within-size behavioral change.

Panel B of Appendix Table B.8 compares wages and output on a per-worker basis, as in the model. QCEW annual pay per employee, deflated by the nonfarm business output deflator, grows by 20.6 log points over 2001–2019. BLS nonfarm business output per worker grows by 31.0 log points, leaving a gap of -10.4 log points.

For a closer model counterpart, we use BLS compensation, employment, and output for the same nonfarm business sector. Let C denote nominal labor compensation including employer benefits, N employment, Y real output, and P the output deflator. Real compensation per worker is $C/(PN)$ and output per worker is Y/N, so

$$\Delta \log\left(\frac{C}{PN}\right) - \Delta \log\left(\frac{Y}{N}\right) = \Delta \log\left(\frac{C}{PY}\right).$$

Real compensation per worker grows by 19.9 log points, giving a compensation–productivity gap of -11.1 log points, equal to the log change in the BLS labor-share index.[32] Employment cancels from this ratio, so aggregate employment growth alone does not mechanically generate the gap. This BLS measure includes imputed self-employed compensation and covers nonfarm business. The BEA employee-compensation measure remains the benchmark for the paper's quantitative contribution.

Panel C applies the same accounting to the model, where the wage is real compensation per worker and the labor share is $w/(Y/L)$. The counterfactual holds the productivity process fixed to isolate the effect of higher adjustment costs. Its wage change therefore excludes productivity

[31] See the BLS BED technical notes.

[32] In the frozen BLS annual series, the labor-share index (2017 = 100) falls from 111.689 in 2001 to 99.995 in 2019, a decline of 11.1 log points. The corresponding labor-share levels are 63.1% and 56.5%, a decline of 6.6 percentage points.

growth. We compare the gap between wage growth and output-per-worker growth with the data. Higher adjustment costs reduce the equilibrium wage by 4.71 log points and output per worker by 3.04 log points, leaving a gap of -1.68 log points. The model reproduces the direction of this gap in the QCEW comparison and matched BLS data, with other forces needed to explain the remainder.

Appendix C. Computational Details

This appendix describes the numerical methods used to solve and calibrate the quarterly model presented in Section 5. The model is solved at a quarterly frequency, includes an i.i.d. transitory shock alongside the persistent productivity process, allows the persistent innovation to be leptokurtic, and constructs all BED- and BDS-convention moments from simulated establishment histories at the observation dates used in the data.

Appendix C.1. State Space and Discretization

Persistent productivity process. The persistent idiosyncratic shock follows an AR(1) process in logs:

$$\log s_t = a + \rho \log s_{t-1} + \eta_t.$$

We discretize this process using the Tauchen (1986) method with $N_s = 100$ grid points on the log-productivity space. The grid spans ± 4 unconditional standard deviations around the unconditional mean $\mu_s = a/(1-\rho)$, with equal spacing Δs (where s_i denotes points on the log grid and the production function uses exponentiated levels e^{s_i}).

The innovation η_t is drawn from a *variance-preserving mixture of normals*. With probability p it is drawn from a wide component and with probability $1-p$ from a narrow component:

$$\eta_t \sim (1-p)\,N\big(0,\sigma_1^2\big) + p\,N\big(0,(\xi\,\sigma_1)^2\big), \qquad \sigma_1^2 = \frac{\sigma_\eta^2}{(1-p)+p\,\xi^2}, \tag{C.1}$$

so that the total variance is $\mathrm{Var}(\eta_t) = \sigma_\eta^2$ for any (p,ξ). The pair (p,ξ) changes the shape of the innovation distribution, including its kurtosis, while leaving persistence ρ and unconditional variance unchanged. The excess kurtosis is

$$\kappa = 3\left[\frac{(1-p)+p\,\xi^4}{[(1-p)+p\,\xi^2]^2} - 1\right],$$

which is zero at $p=0$, $p=1$, or $\xi=1$ (the Gaussian cases). For fixed $\xi > 1$, it increases with p up to $p = 1/(1+\xi^2)$ and decreases thereafter. At the benchmark $(p,\xi) = (0.0183, 11.786)$, a larger jump probability locally reduces excess kurtosis. For fixed $0 < p < 1$, increasing $\xi > 1$ raises

excess kurtosis. The transition matrix $\Pi(s'|s)$ integrates the mixture density over the Tauchen cells. At $p = 0$ or $\xi = 1$, it reduces to the single-normal Tauchen matrix.

Transitory shock. Output also depends on an i.i.d. transitory shock e^{ε_t} that is mean-one, $\mathbb{E}[e^{\varepsilon_t}] = 1$, with $\varepsilon_t \sim N(-\frac{1}{2}\sigma_\varepsilon^2, \sigma_\varepsilon^2)$. We integrate over ε_t using Gauss–Hermite quadrature with $N_\varepsilon = 7$ nodes. Because the transitory shock is i.i.d., it enters the within-period payoff but not the continuation value, so the value function retains the two-dimensional state (s,n) while the flow payoff is evaluated at (s,ε,n).

Entrant productivity. The entrant distribution assigns equal probability to the first 66 of the 100 equally spaced log-productivity nodes and zero to the remaining nodes. It is uniform over these log-grid nodes, rather than productivity levels. Each supported node has probability $1/66$. In the smaller-shock counterfactual, we retain the baseline grid and entrant support and adjust the drift to preserve unconditional mean productivity in levels.

Employment grid. The employment grid uses integer points $n = 1,2,\ldots,300$, followed by a geometric tail with 2% spacing up to $n_{\max} = 5{,}000$, for 440 points in total. Integer resolution permits changes of whole jobs among small establishments, while the geometric tail permits small proportional changes among larger establishments. Appendix Table D.3 reports recalibration and counterfactual checks with 730 employment points, a denser productivity grid, and more transitory-shock nodes.

Exit value. With probability x_0, the exit value is zero. Otherwise it is drawn uniformly from 200 equally spaced points on $[0,2{,}000]$, including zero. The total mass at zero is therefore $x_0 + (1 - x_0)/200$. The exit value represents the value recovered when an establishment closes. Appendix Table D.3 reports alternatives that change the shape of the non-point component while preserving its mean of 1,000.

Appendix C.2. Value Function Iteration

Given a wage w, we solve the firm's dynamic programming problem by iterating on the integrated value function $W(s,n)$ until convergence. The flow payoff includes the transitory shock e^{ε} and the convex adjustment-cost term $\frac{1}{2}\phi(n' - n)^2$, and the continuation integrates over both the persistent transition and the transitory shock. Each iteration proceeds as follows:

1. **Adjustment value.** For each (s,ε,n), compute the value of adjusting to each possible n':

$$V^a(s,\varepsilon,n) = \max_{n'} \left\{ s e^{\varepsilon} (n')^{\theta} - w n' - h(n',n) + \beta W(s,n') \right\}$$

2. **Inaction value.** Compute the value of maintaining current employment:

$$V^n(s,\varepsilon,n) = s e^{\varepsilon} n^{\theta} - wn + \beta W(s,n)$$

3. **Continuation value.** The firm adjusts if $V^a > V^n$ and the optimal $n' \neq n$. The resulting value is $V^c(s,\varepsilon,n) = \max\{V^a(s,\varepsilon,n), V^n(s,\varepsilon,n)\}$.
4. **Expected continuation.** Compute $Z(s,n) = \sum_{s'} \Pi(s'|s) \sum_j \omega_j V^c(s', \varepsilon_j, n)$, where $\{\varepsilon_j, \omega_j\}$ are the Gauss–Hermite nodes and weights for the transitory shock.
5. **Exit decision.** At the beginning of the next period, before observing the new productivity shocks, the firm draws an exit value x and exits if the net exit payoff $x - \gamma n$ exceeds the continuation value $Z(s,n)$. The integrated value is:

$$W(s,n) = \sum_x \Psi(x) \cdot \max\{x - \gamma n,\, Z(s,n)\}$$

   where Ψ is the discretized exit value distribution and γn is the per-worker dismissal cost for releasing the firm's workforce (the convex component ϕ is not charged on exit, so exit severance is purely per-worker).

Value-function iteration uses an absolute sup-norm tolerance of 10^{-8} and a cap of 300 outer iterations. Howard policy iteration accelerates evaluation of a fixed policy. Final parameter choices are checked from a cold start to verify that the reported equilibrium does not depend on the preceding wage or parameter evaluation.

Appendix C.3. Stationary Distribution

Let $n^I(s,\varepsilon,n)$ and $n^e(s,\varepsilon)$ denote incumbent and entrant employment choices, and let $\chi(s,n')$ be the exit probability before the next productivity draws. Per unit of entrant mass, the stationary incumbent distribution satisfies

$$\begin{aligned}\mu^I(s',n') = &\sum_{s,n,\varepsilon} \mu^I(s,n)\, \omega(\varepsilon) \mathbb{1}\{n^I(s,\varepsilon,n) = n'\}[1 - \chi(s,n')]\Pi(s'|s) \\ &+ \sum_{s,\varepsilon} \nu(s)\, \omega(\varepsilon) \mathbb{1}\{n^e(s,\varepsilon) = n'\}[1 - \chi(s,n')]\Pi(s'|s).\end{aligned}$$

We iterate this law of motion until the largest change in any element of the distribution is below 10^{-6}. We apply the employment policy separately for each transitory-shock realization and sum the resulting contributions using the shock probabilities.

Before applying the entrant policy $n^e(s,\varepsilon)$, the code temporarily stores entrant mass in the first employment-grid column of the distribution array. This storage step does not mean that entrants already employ one worker. The entrant policy determines their initial workforce without charging

a labor adjustment cost. In the simulated BED panel, each entrant is recorded as an opening, so its entire initial workforce counts as gross job creation.

Appendix C.4. General Equilibrium

The equilibrium wage w^* satisfies $V^e(w^*) = c_e$. We bracket the root and use a safeguarded secant method, with bisection when needed, until the relative free-entry residual is below 10^{-8}. Each wage evaluation solves the firm problem. Warm starts accelerate successive evaluations, and the final equilibrium is re-evaluated from a cold start.

Appendix C.5. Simulation and Time Aggregation

We simulate quarterly establishment histories at the equilibrium wage and construct the BED and BDS moments at their respective frequencies. Quarterly no-change and adjustment sizes use consecutive-quarter employment. Annual reallocation, entry, and exit use March-to-March observations.

Panel construction. We simulate 100,000 establishments for 120 burn-in quarters followed by 48 measured quarters. Common random numbers are held fixed across parameter evaluations and counterfactuals. Appendix Table D.3 reports checks with a different seed and a panel three times larger, including recalibration and retargeting.

BED conventions. The BED compares employment in the third month of consecutive quarters. Quarterly no-change counts continuers with equal employment and divides by the union of establishments present at either endpoint. Openings and closings are changing establishments. Gains-size shares include openings in both model and data. The calibration uses 1–4, 20+, and nested 50+ shares among gainers. The counterfactual table reports the mutually exclusive 1–4, 5–19, 20–49, and 50+ bins, with 20–49 obtained by subtracting 50+ from 20+. Quarterly gross gains and losses include openings and closings and use the Davis–Haltiwanger–Schuh average-endpoint employment denominator. Employment-weighted inaction uses that denominator as well. Its data counterpart uses BED employment reconstructed from published NSA levels and rates. The gross-flow target is the calendar-year average of seasonally adjusted quarterly rates.

BDS conventions. The BDS measures job flows March-to-March. We aggregate the quarterly panel to annual frequency using point-in-time March snapshots: establishments that enter *and* exit within the year never appear, entrant size is measured at the first March after entry (not at birth), and job creation and destruction follow the standard birth/death and expansion/contraction decomposition. These conventions determine the timing of the simulated annual moments.

Appendix C.6. Calibration Procedure

The benchmark economy is calibrated to ten 2001 moments. With the persistent process set externally as described in Section 5.3, the parameters $(\theta, c_e, a, x_0, \gamma_0, \phi_0, F_0, \sigma_\varepsilon, p, \xi)$ minimize a weighted sum of squared relative deviations:

$$\min_{\Theta} \sum_{j=1}^{10} \omega_j \left(\frac{m_j^{\text{model}}(\Theta) - m_j^{\text{data}}}{m_j^{\text{data}}} \right)^2 .$$

The targets are listed in Panel A of Table 6. The weights are 50 for the labor share, 35 for annual reallocation, 5 for quarterly count-based inaction, and 1 for each remaining moment, including quarterly gross flows and employment-weighted inaction.

The conditional 1–4 and 20+ shares describe small and large adjustments, while the nested 50+ share adds information about the upper tail. Employment-weighted inaction describes the employment share at establishments that do not adjust. Nelder–Mead searches use common simulated draws and box bounds on the parameters. A search stops after an improvement of less than 10^{-4} over 120 evaluations or at its evaluation budget. The selected parameter vectors are checked from cold starts. Because employment policies and adjustment bins are discrete, the objective is piecewise and nonsmooth. The selected loss is 0.09594.

For the baseline counterfactual, we search over nondecreasing kink and quadratic cost parameters while keeping the fixed cost and all other parameters unchanged. We target changes of $+6.5085$ percentage points in count-based inaction and -3.5250 points in quarterly gross flows relative to the model's own benchmark. Each target must be matched within 0.1 percentage point. The selected multipliers are $\gamma_1/\gamma_0 = 1.5385$ and $\phi_1/\phi_0 = 2.9399$, with target residuals of $+0.0818$ and -0.0321 points. The final endpoint is checked from a cold start.

Appendix C.7. No-Change and Size-of-Change Measurement

The quarterly no-change rate and the conditional size-of-change shares are the moments that discipline the friction. They are measured *directly* from the simulated panel, following the BED conventions described in Appendix C.5, which removes the need for a latent-inaction proxy.

An establishment contributes to the quarterly no-change count if it is present at both endpoints with unchanged employment. The denominator is the union of establishments present at either endpoint. Conditional gains-size shares include openings and use the same population as the BED data. Employment-weighted no-change divides employment at unchanged establishments by total average-endpoint employment. These definitions are applied to the common simulated panel used for gross-flow and annual reallocation moments.

The employment grid includes every integer through 300 workers, so a one-worker change is feasible among small establishments. The discrete policies leave a piecewise objective. The selected

Appendix Table C.1: Sensitivity: Moment Elasticities with Respect to Calibrated Parameters

Panel A: Aggregate moments and turnover

Parameter	Wage	Labor share	Annual JRR	Entry	Size
θ	4.544	0.932	1.393	-0.082	-1.641
c_e	-4.803	-0.559	-3.941	-5.794	13.649
a	1.869	0.074	0.666	-0.054	-1.683
x_0	-2.886	-0.278	-2.626	-6.694	7.523
γ_0	-0.002	0.000	-0.010	0.002	0.005
ϕ_0	-0.039	-0.012	-0.196	0.003	0.013
σ_ε	0.016	-0.016	0.141	-0.002	-0.038
p	-0.019	0.006	-0.020	0.015	0.029
ξ	0.070	-0.016	-0.019	0.014	-0.197
F	0.000	0.000	-0.001	-0.001	0.002

Panel B: Quarterly adjustment moments

Parameter	No-change	Gross flows	1–4	20+	50+	Weighted NC
θ	0.633	2.191	-0.303	-0.298	3.029	-1.610
c_e	-4.476	-6.060	-0.840	9.341	10.877	-2.511
a	0.027	1.327	-0.006	-1.567	-1.352	-0.448
x_0	-2.442	-3.763	-0.363	4.500	6.256	-0.597
γ_0	0.048	-0.037	-0.004	0.048	0.064	0.199
ϕ_0	0.062	-0.369	0.139	-0.168	-0.489	0.234
σ_ε	-0.327	0.422	-0.149	-0.056	-0.182	-0.485
p	0.000	-0.038	0.019	0.151	0.040	0.007
ξ	0.060	0.038	0.086	0.043	0.492	0.209
F	0.002	-0.003	-0.004	0.004	0.005	0.013

Notes: Elasticities use symmetric ±5% parameter perturbations with the equilibrium wage re-solved at each point. Panel B shares are conditional on gains, including openings. Weighted NC is employment-weighted no-change. The entries describe local responses around the benchmark.

counterfactual matches both quarterly changes within 0.1 percentage point. Recalibrations with finer grids and alternative simulation settings are reported in Appendix D.3.

Appendix C.8. Sensitivity Analysis

Appendix Table C.1 reports local elasticities of all ten calibration moments and the wage with respect to the ten calibrated parameters, using ±5% perturbations. The quadratic cost lowers gross flows and the far-tail share while raising the small-adjustment share. Both the kink and quadratic costs raise employment-weighted inaction. The two mixture parameters have different tail responses, with the jump size having its largest reported adjustment-share effect on 50+. The fixed cost has small local effects at the benchmark. Larger increases affect inaction and adjustment

sizes, as illustrated in Appendix Figures D.1 and D.2. Discrete employment policies can make these finite-difference responses sensitive to the perturbation size. These derivatives describe local information in the moments, rather than establish unique identification. The recalibrations in Appendix Table D.3 supplement the local comparisons.

Appendix D. Model Robustness Appendix

Appendix D.1. Employment Flows by Adjustment Size

Appendix Table D.1 reports the employment-flow shares discussed in Section 6. The shares weight employment changes by the number of jobs and cover gains and losses at continuing, opening, and closing establishments.

Appendix Table D.1: Employment Flows by Adjustment Size

Jobs adjusted	Model 2001	Model CF	Model Δ	Data 2001	Data 2019	Data Δ
1–4	39.54	46.84	+7.30	30.13	38.15	+8.02
5–19	36.45	33.40	-3.05	28.17	28.81	+0.63
20+	24.02	19.76	-4.25	41.70	33.05	-8.65

Notes: Entries are shares of total gross job gains plus losses, weighted by jobs adjusted. The model and BED series include openings and closings. Changes are percentage points. These job-weighted shares differ from the establishment-count shares conditional on gains in Table 7. The data series available for this comparison groups changes of 20 or more jobs together.

Appendix D.2. Alternative Cost Increases and Targets

The exercises in this subsection start from the same ten-moment calibration and hold all parameters other than the indicated adjustment costs fixed. Appendix Table D.2 compares the baseline and data with cost increases targeted to annual reallocation. The labor share is untargeted throughout.

Targeting annual reallocation. The main regressions in Section 3 relate annual reallocation to labor shares. This relationship motivates an alternative counterfactual that targets the full annual reallocation decline. The bridge regressions in Section 4.2 connect inaction to declines in annual reallocation and payroll shares, providing evidence on the adjustment mechanism. Comparing the two targeting choices shows how the model's contribution changes when quarterly adjustment behavior also constrains the cost increase. The data and baseline rows in Appendix Table D.2 make this comparison explicit.

To match the full annual reallocation decline, the kink and quadratic parameters rise proportionally by a factor of 4.81. Annual reallocation falls by 8.54 points, within 0.1 point of

Appendix Table D.2: Targeting the Annual Reallocation Decline

Exercise / cost parameters raised	Δ NC	Δ annual JRR	Δ LS	Δ 1–4	Δ 50+	Δ gross flows	LS contribution (%)	JRR contribution (%)	Annual JRR target matched
Data, 2001–2019	+6.51	-8.61	-3.94	+3.07	-0.29	-3.53	—	—	—
Baseline (quarterly targets)	+6.59	-5.88	-0.95	+4.86	-0.43	-3.56	24.1	68.3	—
Kink + quadratic	+14.09	-8.54	-1.47	+3.16	-0.60	-4.92	37.2	99.2	Yes
Quadratic	+9.88	-8.67	-1.53	+5.04	-0.68	-4.81	38.9	100.7	Yes
Kink	+33.85	-8.63	-1.12	-10.75	+1.47	-5.37	28.4	100.3	Yes
Fixed	+29.17	-1.43	+0.03	-23.05	+1.00	-2.34	-0.6	16.6	No

Notes: All model exercises use the common 2001 benchmark. NC denotes quarterly no-change. The baseline raises the kink and quadratic cost parameters by independently chosen factors to match the changes in quarterly no-change and gross flows. The four rows below the baseline instead target the 8.606-point decline in annual reallocation. The kink-plus-quadratic row raises both parameters by a common factor. The other annual-target rows raise one parameter. All remaining cost parameters stay at their positive benchmark values. The last column records whether the annual target is met within 0.1 percentage point. The fixed-cost row reports the best result found in the search. The labor share is untargeted in every model exercise. Size shares are conditional on gains, including openings. Changes are percentage points. Contributions are percentages of the observed labor-share and annual reallocation declines.

the observed decline, and the labor share falls by 1.47 points, or 37.2%. The small-adjustment share rises by 3.16 points, close to the observed 3.07-point increase. Inaction, however, rises by 14.09 points and quarterly gross flows fall by 4.92 points, compared with 6.51 and 3.53 in the data. The larger labor-share contribution thus comes with more inaction than the data suggest. This difference supports using the quarterly targets to discipline the frequency and total size of employment adjustments in the baseline. Together, the baseline contribution of 24.1% and this annual-target contribution give the 24–37% comparison in the main text. Their annual reallocation contributions are 68.3% and 99.2%, with annual reallocation untargeted in the baseline and targeted in the alternative.

Continuing adjustment costs absorb 4.1% of the wage bill in the benchmark, 5.0% in the baseline counterfactual, and 5.5% in the proportional annual-reallocation exercise. Expenditure rises despite less adjustment because it reflects both the cost schedule and the employment policies it induces.

Individual cost increases. The remaining rows in Appendix Table D.2 target the same annual reallocation decline by increasing one cost parameter at a time. The quadratic cost alone gives a labor-share contribution of 38.9% and shifts gains toward smaller sizes, but also raises inaction more than in the data. The kink alone reaches the annual target only with a much larger increase and lowers the small-adjustment share. The fixed-cost search reaches its upper multiplier of 100 without matching the annual decline. These comparisons show why the aggregate flow decline and the adjustment pattern need to be considered together. The 24–37% comparison above uses the two specified kink-plus-quadratic exercises.

Appendix Table D.3: Calibration and Counterfactual Robustness

Specification	Loss	Δ NC	Δ gross flows	LS contribution	JRR contribution
Benchmark	0.0959	+6.59	-3.56	24.1	68.3
Panel A: Asymmetric hiring and firing costs					
Firing kink / hiring kink = 4	0.0946	+6.50	-3.55	24.0	68.1
Firing kink / hiring kink = 1/4	0.0963	+6.54	-3.57	24.1	68.4
Panel B: Exit-value distribution					
Exit-value shape 1	0.1310	+6.50	-3.54	25.3	68.6
Exit-value shape 2	0.1416	+6.51	-3.59	24.5	68.8
Panel C: Adjustment costs at exit					
Half quadratic exit cost	0.0967	+6.58	-3.56	30.0	69.0
Full quadratic exit cost	0.0970	+6.57	-3.61	33.0	70.5
Panel D: Simulation settings					
Alternative simulation seed	0.1016	+6.53	-3.62	25.2	70.9
300,000 simulated establishments	0.0987	+6.53	-3.62	24.9	70.4
Panel E: Grid resolution					
11 transitory-shock nodes	0.0946	+6.43	-3.56	24.5	68.6
199 persistent-shock nodes	0.0926	+6.49	-3.54	24.6	68.3
730 employment-grid points	0.1020	+6.54	-3.58	24.4	69.2
Panel F: Proportional gross-flow targeting					
Proportional gross-flow decline	0.0959	+6.44	-2.52	13.3	42.2

Notes: Each alternative in Panels A–E recalibrates the 2001 economy to the same ten moments and then re-targets the baseline changes using the kink and quadratic parameters. Panel F holds the benchmark calibration fixed and chooses the kink and quadratic parameters to match the same count-based inaction change and the proportional gross-flow decline. The gross-flow target applies the data's proportional decline to the model's initial gross-flow rate. All reported counterfactuals meet their respective two targets within 0.1 percentage point. Contributions are percentages of the observed labor-share and annual reallocation declines. Exit-value shapes preserve the conditional mean. Numerical variants change one simulation or grid setting at a time. Loss is the common weighted relative-moment objective. Panel F reports the unchanged benchmark calibration loss. NC denotes count-based inaction. Changes in NC and gross flows are percentage points.

Appendix D.3. Additional Robustness

Appendix Table D.3 examines cost asymmetry, exit assumptions, numerical settings, and proportional gross-flow targeting. Panels A–E recalibrate the 2001 economy to the same ten moments and then choose increases in the kink and quadratic parameters to match the baseline changes in quarterly inaction and gross flows. Panel F holds the benchmark calibration fixed and changes the gross-flow target. Each reported counterfactual matches its two targets within 0.1 percentage point.

Asymmetric hiring and firing costs. In Panel A, we set the firing-to-hiring ratio of the kinked per-worker cost to four or one quarter and keep that ratio fixed in the counterfactual. The overall fit and the quarterly gaining/losing split change little. Both specifications give contributions of about 24% of the labor-share decline and 68% of the annual reallocation decline, close to the benchmark.

Exit-value distribution. In Panel B, the rows labeled "Exit-value shape 1" and "Exit-value shape 2" replace the uniform component of the exit-value distribution with two decreasing densities, rescaling their support to preserve its mean. Both place more weight on low exit values and increase dispersion. Their calibration losses are higher, but the labor-share and annual reallocation contributions remain close to the benchmark, at about 25% and 69%.

Adjustment costs at exit. In the benchmark, exiting establishments pay the per-worker dismissal cost, while the fixed and quadratic adjustment costs apply to continuing establishments. The half and full quadratic exit-cost rows in Panel C add $\phi n^2/4$ and $\phi n^2/2$, respectively, to the dismissal cost when an establishment with n workers exits. After recalibration, these treatments give labor-share contributions of 30% and 33%, compared with 24% in the baseline. The annual reallocation contribution remains between 69% and 71%. Thus exit costs affect the magnitude of the labor-share response.

Simulation settings. In Panel D, we change the simulation seed or increase the panel from 100,000 to 300,000 establishments. After recalibration and matching the two quarterly changes, the labor-share contributions are 25.2% and 24.9%, respectively. The annual reallocation contributions are 70.9% and 70.4%. These results are close to the benchmark.

Grid resolution. Panel E increases the number of transitory-shock nodes from 7 to 11, persistent-productivity nodes from 100 to 199, or employment-grid points from 440 to 730. Each exercise changes one grid at a time. After recalibration and matching the two quarterly changes, the labor-share contributions range from 24.4% to 24.6% and the annual reallocation contributions from 68.3% to 69.2%. The aggregate results are stable across these refinements.

Proportional gross-flow targeting. In Panel F of Appendix Table D.3, we hold the 2001 calibration fixed and choose the kink and quadratic cost increases to match the change in count-based inaction and the proportional decline in quarterly gross flows. We apply the observed proportional decline to the model's initial gross-flow rate, which is below its empirical counterpart. Gross flows fall by 2.52 percentage points, compared with 3.56 in the baseline. The exercise accounts for 13.3% of the labor-share decline and 42.2% of the annual reallocation decline, compared with 24.1% and 68.3% in the baseline. Thus the quantitative contribution depends on how the gross-flow decline is targeted.

Appendix D.4. Decomposition of the Labor-Share Decline

This subsection gives the formulas behind Panel C of Table 7 and further detail on its two components. The model labor share is $\text{LS} = wL/Y$, where w is the wage, L is employment, and $Y = \sum_i f(s_i, \varepsilon_i, n_i)$ is gross output. Adjustment costs use up resources, but gross output is measured before them. These costs therefore affect the labor share only through the wage and employment per unit of output, $\ell = L/Y$. Employment and output are measured per unit of entrant mass, following the normalization in Section 5. The entrant mass scales both together and cancels from the labor share. Subscripts 0 and 1 denote the benchmark and counterfactual equilibria.

The Shapley decomposition splits $\Delta\text{LS} = w_1\ell_1 - w_0\ell_0$ into a wage component and an allocation component. If the wage changes first, the wage contributes $(w_1 - w_0)\ell_0$ and the change in ℓ contributes $w_1(\ell_1 - \ell_0)$. If ℓ changes first, its change contributes $w_0(\ell_1 - \ell_0)$ and the wage contributes $(w_1 - w_0)\ell_1$. Averaging the two orders assigns half of the interaction $(w_1 - w_0)(\ell_1 - \ell_0)$ to each component:

$$\Delta\text{LS} = \underbrace{\tfrac{1}{2}(w_1 - w_0)(\ell_0 + \ell_1)}_{\text{wage component}} + \underbrace{\tfrac{1}{2}(w_0 + w_1)(\ell_1 - \ell_0)}_{\text{allocation component}}.$$

The two terms sum exactly to ΔLS. In the baseline, the wage component is -2.67 points and the allocation component is $+1.72$ points, for a total of -0.95 points.

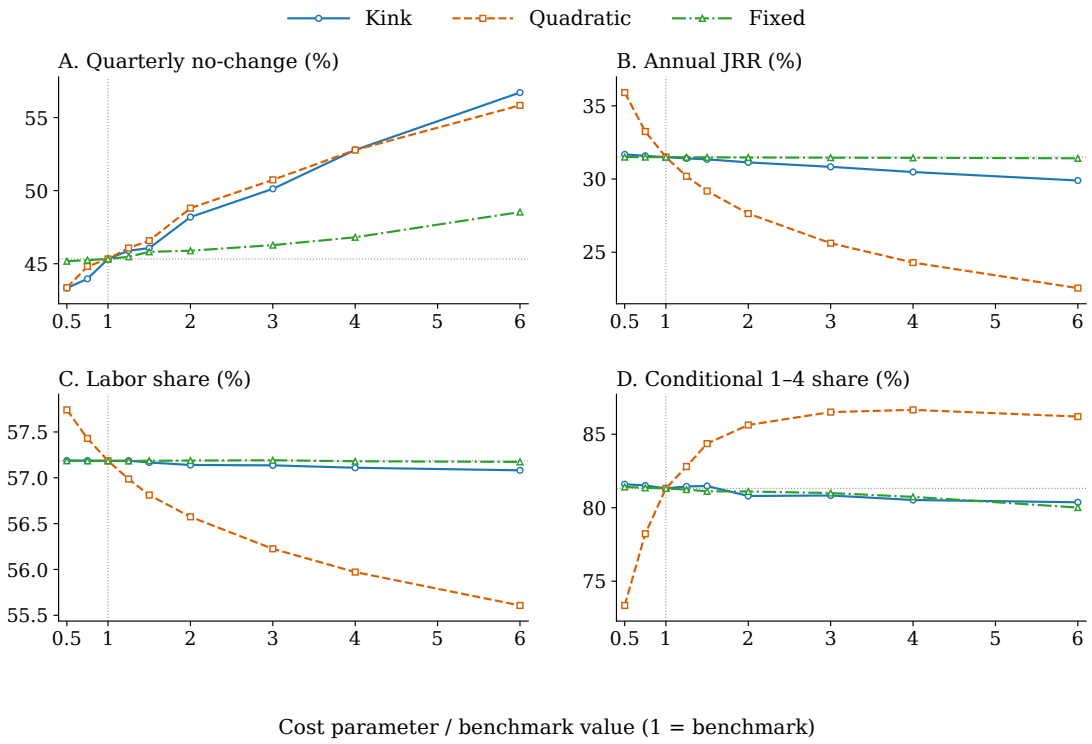
The standard log decomposition also holds exactly: $\Delta \log \text{LS} = \Delta \log w + \Delta \log L - \Delta \log Y$. Multiplying by 100 expresses each change in log points. The wage contributes -4.71 and the rise in L/Y contributes $+3.04$. The labor share therefore falls by 1.68 log points. The fall in output per worker partly offsets the wage contribution.

Appendix Table D.4: Entry Value and the Equilibrium Wage Response

Moment	Value
Entry-value elasticity with respect to the wage	-0.230
Entry-value change at the initial wage (log points)	-1.00
Wage change implied by the local elasticity (log points)	-4.34
Equilibrium wage change (log points)	-4.71

Notes: The entry-value elasticity ζ uses symmetric one-percent wage perturbations at the benchmark. The fixed-wage comparison applies the counterfactual costs while holding the wage at its initial value and re-solving establishment policies. The local wage prediction is $-\Delta \log V^e|_{w_0} / \zeta$.

The wage component reflects the decline required by free entry. Appendix Table D.4 shows a decomposition of the equilibrium wage response. The equilibrium wage falls from 0.990 to 0.945. At the benchmark wage, higher costs lower entry value by 1.00 log points. The local



Appendix Figure D.1: Comparative Statics: The Distinct Roles of the Kink, the Curvature, and a Fixed Cost

Notes: Each panel compares the kink, quadratic, and fixed cost paths on a common vertical scale. The horizontal axis is the varied cost parameter divided by its positive benchmark value. Each path varies one parameter from 0.5 to 6 times its benchmark, holding all other parameters fixed and re-solving the equilibrium wage. Outcomes are percentages. The 1–4 share is conditional on gains, including openings. The dotted vertical and horizontal lines mark the benchmark parameter and outcome, respectively.

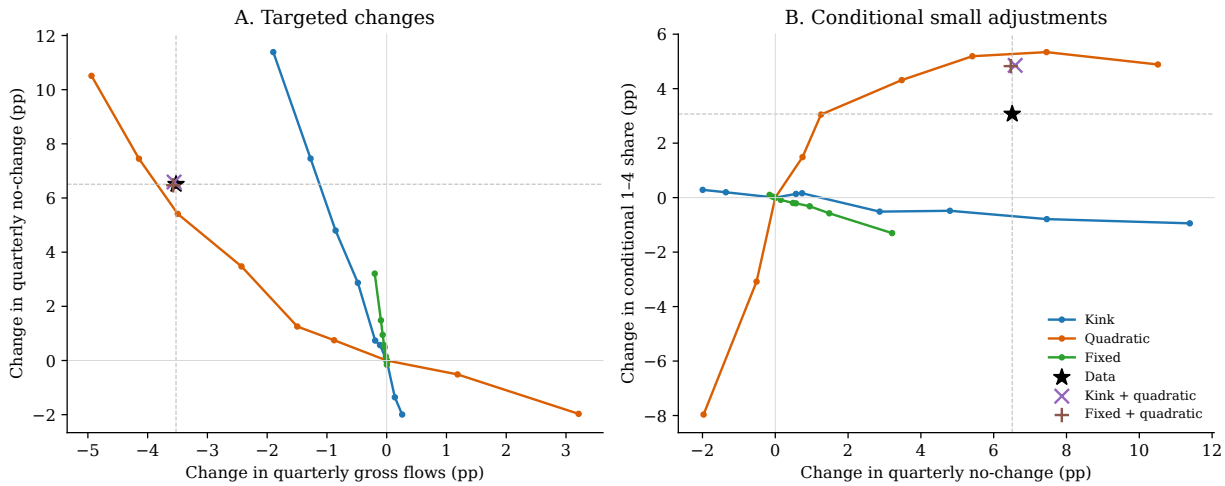
elasticity of entry value with respect to the wage is $\zeta = -0.230$. The first-order approximation $\Delta \log w \approx -\Delta \log V^e|_{w_0} / \zeta$ predicts a wage decline of 4.34 log points, or 92% of the response.

The allocation component captures the change in employment per unit of output. It reflects inaction, adjustment sizes, endogenous exit, and the employment response to the wage. Per unit of entrant mass, employment rises by 1.0% and output falls by 2.0%. The annual entry rate remains about 10.3%, so establishment turnover changes little.

Appendix D.5. The Roles of the Cost Components

This subsection brings together the component comparisons used in Section 6.2.

Component comparative statics. Appendix Figure D.1 separates the effects of the three cost components by varying one cost parameter between 0.5 and 6 times its positive benchmark value while holding the remaining parameters fixed. For the same proportional increase from the benchmark, the quadratic term has much larger effects on annual reallocation and the labor share than the kink or fixed cost. The fixed cost has little effect on these aggregates over the plotted



Appendix Figure D.2: Adjustment Costs and the Two Targeted Margins

Notes: Each line varies one cost parameter between 0.5 and 6 times its positive benchmark value, holding the remaining parameters fixed. Panel A plots the changes in quarterly gross flows and count inaction. Panel B plots inaction and the conditional 1–4 share. The star marks the data, the cross the baseline kink-plus-quadratic counterfactual, and the plus the fixed-plus-quadratic counterfactual. Both joint increases match the two quarterly targets.

range, although it raises inaction and lowers the conditional small-adjustment share. Raising the kink makes each job added or cut more costly. Raising the fixed cost makes an adjustment event more costly regardless of its size, encouraging establishments to wait and combine changes. Raising the quadratic cost makes larger changes disproportionately costly and shifts gains toward smaller sizes. All three terms affect the smallest adjustment, since a one-job change costs $F + \gamma + \phi/2$. Their implications for larger changes differ because only the quadratic term raises marginal cost with adjustment size.

Joint quarterly changes. Panel A of Appendix Figure D.2 plots changes in gross flows against changes in count inaction by varying one cost parameter between 0.5 and 6 times its positive benchmark value while holding the remaining parameters fixed. The quadratic-cost path shows a larger decline in gross flows relative to its rise in inaction than the kink and fixed-cost paths. The joint kink-plus-quadratic and fixed-plus-quadratic increases both reach the two quarterly targets. Panel B adds the conditional small-adjustment share, which rises with the quadratic cost and falls with the fixed cost. This untargeted response helps distinguish cost increases that produce similar inaction.

The figures describe changes from the calibrated economy. Appendix Table C.1 separately reports local elasticities of the 2001 moments with respect to all ten calibrated parameters. Together, these comparisons explain which moments respond to each cost component. They support a quadratic cost increase in the successful two-target exercises, while the two quarterly changes alone

Appendix Table D.5: Alternative Pairs of Rising Adjustment Costs

Cost parameters raised	γ/γ_0	ϕ/ϕ_0	F/F_0	Δ NC	Δ gross flows	Δ annual JRR	Δ LS	LS contribution (%)	JRR contribution (%)	Matched
Kink + quadratic	1.54	2.94	1.00	+6.59	-3.56	-5.88	-0.95	24.1	68.3	Yes
Fixed + quadratic	1.00	3.04	2.64	+6.47	-3.57	-5.95	-0.97	24.7	69.2	Yes
Kink + fixed	4.16	1.00	1.03	+7.74	-1.32	-1.06	-0.07	1.9	12.3	No

Notes: Each row starts from the same ten-moment benchmark and raises the indicated pair of cost parameters. The third cost parameter remains at its calibrated positive value. The targets are a 6.509-point increase in quarterly no-change and a 3.525-point decline in quarterly gross flows. Matched means that both residuals are within 0.1 percentage point. The kink-plus-fixed row reports the best result found in the search and does not match both targets. Changes are percentage points. Contributions are percentages of the observed labor-share and annual reallocation declines.

do not select which other cost parameter rises. Appendix Tables D.2 and D.5 report the targeted searches, some of which require increases beyond the range displayed in the figures.

Appendix D.6. Alternative pairs of cost increases under the quarterly targets

Appendix Table D.5 compares the three pairs of cost increases under the changes in quarterly count inaction and gross flows. The third cost parameter remains at its positive calibrated value. Kink-plus-quadratic and fixed-plus-quadratic increases both match the targets, giving labor-share contributions of 24.1% and 24.7% and annual reallocation contributions of 68.3% and 69.2%. Thus the successful pairs give a narrow contribution range under common targets. The completed kink-plus-fixed search does not match both targets. A quadratic cost increase is common to the two successful comparisons, while either the kink or the fixed cost can accompany it.

Appendix D.7. Rising Frictions versus Declining Shock Dispersion

Decker et al. (2020) study employment growth conditional on productivity and initial employment. They find that employment has become less responsive and relate this change to declining reallocation and a drag on aggregate productivity. Their adjustment-cost framework motivates our comparison. We ask how the distinction between smaller shocks and weaker responses extends to the labor share when establishment decisions and the equilibrium wage are determined jointly.

We compare rising adjustment costs with declining productivity-shock dispersion under the same two targets: the rise in quarterly no-change and the decline in quarterly gross flows. The cost exercise varies (γ, ϕ). The shock exercise varies the transitory and persistent innovation dispersions $(\sigma_\varepsilon, \sigma_\eta)$. Both start from the same 2001 calibration. The shock experiment preserves unconditional mean productivity in levels and holds the productivity support, including entrant support, fixed. The equilibrium wage is solved separately in each exercise. Appendix Table D.6 reports the targeted and untargeted changes.

Appendix Table D.6: Rising Frictions versus Declining Shock Dispersion

	No-change	Gross flows	Annual JRR	Labor share
Rising costs (γ, ϕ)	+6.59	-3.56	-5.88	-0.95
Smaller shocks $(\sigma_\varepsilon, \sigma_\eta)$	+6.61	-3.58	-5.45	+1.25
Data, 2001–2019	+6.51	-3.53	-8.61	-3.94

Notes: Changes are percentage points. Both exercises target the changes in quarterly no-change and gross flows. Annual reallocation and the labor share are untargeted. The shock experiment preserves unconditional mean productivity in levels and holds the productivity support, including entrant support, fixed.

Both mechanisms match the quarterly changes and produce similar declines in annual reallocation, at 5.88 points with higher costs and 5.45 points with smaller shocks. Rising costs lower the labor share by 0.95 points, whereas smaller shocks raise it by 1.25 points. Higher costs lower establishment value at a given wage, and the free-entry wage falls proportionately more than output per worker. With smaller shocks, output per worker falls proportionately more than the wage. The identity $LS = w/(Y/L)$ therefore gives opposite labor-share responses.

The labor share thus provides information beyond the employment-flow responses in this comparison. Lower output per worker alone would raise the labor share at a given wage. The free-entry wage response is therefore essential to extending the responsiveness argument to a joint explanation of declining reallocation and the labor share.